

The Value of Choice in Centralized School Assignment: Welfare Gains from Acting on Family Preferences in Ecuador*

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Abstract

We study a 2021 reform in Manta, Ecuador, that replaced a centralized rule minimizing home-to-school distance with deferred acceptance acting on families' reported rank-ordered preferences. Across the three entry grades, the new rule raises the share of applicants assigned to a listed school from 50% to 78% and to their stated first choice from 42% to 70%, while average travel rises by 0.32 km. Because a single, coordinated, largely digital algorithm cleared the market before and after, what principally changed was the objective, isolating the value of acting on preferences. Assignment rates are not welfare: valuing the additional travel requires a model of preferences, which we estimate from the submitted lists, denominating welfare in kilometres of willingness to travel. In Preschool 1, the least congested grade, deferred acceptance delivers 0.69 km per applicant, net of the 0.33 km of additional travel it induces, or 87.6% of the range to the unconstrained maximum and 99.6% of the narrower, priority-respecting range. The magnitude is what is falsifiable: on the in-market simulation pool rather than the welfare sample, the entry-grade gain is 0.46 km under the estimated preferences but 0.0005 km under a proximity-only null with lists rebuilt at each family's observed length. We measure two inputs usually taken as given: a concave cost in *walking* time predicts held-out rankings 485 log-points better than the near-universal linear straight-line specification, and we derive school characteristics, thin in administrative data, from satellite and street-level imagery. Most of the dispersion in school desirability remains unexplained, which is the argument for eliciting preferences rather than proxying them.

JEL: C78, D47, I20, I21

Key words: Mechanism design, centralized student assignment, school choice, Ecuador

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1 Introduction

Coordinated, digital school-choice systems are spreading rapidly across the developing world (Aguilera et al., 2022; World Bank, 2024; Elacqua et al., 2021), now allocating millions of children and raising a first-order design question: how much does it matter that the system act on what families *want*, instead of simply assigning them to the nearest school? This paper examines the welfare effects of a policy change in Manta, Ecuador’s coordinated school assignment system, which previously aimed to minimize travel distance. That system was costly to operate, requiring accurate georeferencing and case-by-case adjustments for geographic barriers like hills and rivers. The Ecuadorian Ministry of Education, in partnership with the Inter-American Development Bank (IADB) and ConsiliumBots, piloted a new system driven by applicants’ preferences. The new policy followed best practices in the design of centralized assignment, using the deferred acceptance algorithm with unlimited rank-ordered lists (Abdulkadiroğlu & Sönmez, 2003; Pathak, 2011, 2017; Abdulkadiroğlu et al., 2017).

To compare the assignment alternatives we assemble data that go beyond what a proximity rule requires. The system implemented in Manta elicited rank-ordered preferences alongside the georeferenced home location that a distance rule itself needs; to these we add a survey of 1,873 parents fielded before results were published, network travel times by road and on foot, and school characteristics measured from satellite and street-level imagery—administrative records describe these schools with little beyond a handful of facility indicators. We treat the submitted reports as truthful on the strength of the mechanism’s incentive properties (Proposition 1), not as directly observed truth, and we probe both maintained assumptions below. Together these allow us to compare the assignments made by the new centralized choice and assignment system (CCAS) with the simulated assignments of the prior alternative, and to ask what a demand model estimated on this market is in fact fitting. Our methodology replicates the rules of the previous process and simulates assignments under different lotteries following Abdulkadiroğlu et al. (2017).

Our key finding is that switching a centralized algorithm’s objective from proximity to elicited preferences delivers large welfare gains—even though a distance rule looks nearly optimal, since most families rank their nearest school first. The gains are real rather than mechanical: more than a quarter of families (28%) rank as their first choice a school more than 100 metres farther than their nearest, accepting a median of 0.59 km of additional travel to do so (Figure 2), and a distance rule treats this preference as pure cost. Adopting the deferred acceptance (DA) algorithm raises the share of applicants assigned to a school they listed from 50% to 78% and to their stated first choice from 42% to 70%, while average travel distance rises by only 0.32 km. Expressed as the share of the distance rule-to-constrained-efficient range that DA closes, the gain is about 99.6% in Preschool 1. That benchmark respects priorities and is reached from deferred acceptance by Pareto-improving trades, so it is not the unconstrained cardinal maximum. We call this span the *attainable range* throughout—attainable, that is, by a mechanism that respects the system’s priorities—and we quantify its distance to the unconstrained maximum in Section 6.3.

We focus throughout on this entry grade, the earliest point of entry and the least congested; the later-entry grades (Preschool 2 and Primary 1), whose applicants are a selected mid-stream population—families that moved or left a school—are harder to model and interpret, recover

smaller and less precisely estimated gains, and are reported in Appendix B.1. Throughout, the coordination of supply and demand is held fixed; what changes is principally whether the coordinating algorithm allocates on proximity or on preferences. The move also introduces reported-over-imputed priority and school-specific tie-breaking, and our simulated distance benchmark omits the historical school-change petition phase; Table 6 isolates the first two for assignment rates, and we read the evidence as attributing the gain principally, rather than exclusively, to acting on preferences.

Two features sharpen the result. First, in Preschool 2 the gains are *progressive*: they accrue disproportionately to families in lower-schooling neighbourhoods, so there acting on preferences is also equity-enhancing. In Preschool 1 the same endpoint contrast is not distinguishable from zero once resampling respects the census block at which the proxy varies, so we do not read a gradient there. Second, this qualitative lesson—that, where seats are slack, the algorithm matters far less than whether the system acts on preferences—echoes the New York findings despite the vast difference in setting (a huge, wealthy city versus a small coastal town), suggesting it is not a peculiarity of one place.

We focus our analysis on estimated welfare and on the share of applicants assigned to more or less preferred alternatives based on their reported preferences, as detailed in Section 5. We do not consider school quality, for two reasons: i) we aim not to study whether families in Ecuador prefer higher-quality schools, but to assess the welfare consequences of the assignment system as it relates to applicants’ valuation of different schools; and ii) we cannot directly observe school quality.¹

These results contribute to the literature on the welfare effects of centralized school choice. The most closely related study is [Abdulkadiroğlu et al. \(2017\)](#), who estimate the gains from New York City’s switch from an *uncoordinated*, multiple-offer system to a coordinated deferred-acceptance mechanism, and find that most of the gain comes from coordinating offers, not from the choice among coordinated mechanisms. To our knowledge theirs remains the only prior estimate of this welfare object on a comparable scale; the broader empirical literature instead compares coordinated mechanisms to one another ([de Haan et al., 2023](#); [Calsamiglia et al., 2020](#); [Agarwal & Somaini, 2018](#)) or evaluates information provision within an already-coordinated system ([Arteaga et al., 2022](#)); a separate strand quantifies the administrative savings from digitalizing and centralizing the application process itself, relative to an in-person, decentralized one ([Aguilera et al., 2022](#)). Our setting supplies the complementary experiment that no move out of an uncoordinated system can: a single centralized algorithm cleared the Manta market both before and after the reform, so our comparison holds the coordination of supply and demand fixed—and largely the digital application process, though the prior distance procedure retained some in-person steps—and varies only whether the algorithm allocates on proximity or on elicited preferences.

Our contribution is therefore threefold. First, we estimate the welfare gain from acting on household preferences against a benchmark that was *enacted policy*, not a constructed counterfactual: the distance rule we compare deferred acceptance to is the rule the system itself

¹The Ecuadorian government does not currently apply census-based student learning assessments in primary grades. We also do not study the system’s impact on other measures, such as educational segregation: individual socioeconomic records for participating applicants are not available to us, and the census-block schooling proxy we do observe supports an analysis of who gains but not of how students are composed across schools.

had been using, replicated on these applicants, whereas the neighbourhood benchmark in [Abdulkadiroğlu et al. \(2017\)](#) is one the authors construct. And—among the very few welfare evaluations of a move to preference-based centralized assignment, after [Abdulkadiroğlu et al. \(2017\)](#) for New York—ours is the first for a developing country and for entry-level grades, where centralized assignment has spread fastest but been studied least. Second, because coordination and digitization were already in place, the Manta reform isolates *preferences* as the channel: New York’s move *out of* an uncoordinated system bundles the value of coordinating offers, of digitization, and of acting on preferences, and cannot say how much of its estimated gain is the last of these, whereas holding coordination and digitization fixed allows us to attribute our gain principally to preferences. This is the object the literature has long argued for but could not quantify on its own: prior comparisons either vary coordination, digitization, or information at the same time, or—as in the studies that compare coordinated mechanisms to one another—have no arm that ignores preferences altogether. Third, we measure the two inputs on which the welfare estimate rests and which this literature usually takes as given: travel cost, where a concave function of network *walk* time predicts the submitted rankings markedly better than the near-universal specification that is linear in straight-line distance (Section 5.3), and the schools themselves, whose observable characteristics are thin in administrative data and which we measure from satellite and street-level imagery. The imagery recovers the *level* of non-distance desirability, not heterogeneity in how families substitute between schools, and most of that desirability remains unexplained even with every observable we assemble, which is the argument for eliciting preferences, not proxying them.

The gains do not reflect information provision. A companion survey shows that families who ranked few schools overwhelmingly cite the scarcity of acceptable nearby options, not any difficulty finding alternatives (Section 4); their knowledge of the alternatives they did not apply to is nonetheless limited, a qualification we take up in Section 6.4. Information can matter a great deal in centralized systems: [Arteaga et al. \(2022\)](#) show that warning applicants who are at risk of going unassigned leads them to lengthen their lists and improves their placements. But that channel runs through beliefs about admission chances, and it is not what the Manta reform changed: what changed is the data the algorithm acts on, not what families knew. The old system simply did not act on the preferences families already held.

The remainder of the paper proceeds as follows. Section 2 describes the setting and the two assignment mechanisms; Section 3 situates our contribution in the literature; Section 4 presents the pilot data; Section 5 estimates household preferences and sets up the mechanism comparison; Section 6 reports the assignment, stability, and welfare results; and Section 7 discusses.

2 Context and Algorithm Descriptions

We study school assignment in the coastal region of Manta, Ecuador, concentrating on the urban areas within and around the city of Manta, including the geographic units (“*cantones*”) of Manta, Montecristi, and Jaramijó.

Manta was selected through a process that aimed to find a small but representative city to scale up the school assignment policy.² The selection process considered students in the urban

²There was a change in government in Ecuador in 2021, and the new administration recently decided to scale

area, school coverage, distribution of school types (mainly public and private), as well as city size. Ultimately, Manta was chosen for its relative similarity to the alternatives. Table 9 of Appendix B compares the main characteristics of Manta and Guayaquil, another coastal city and the country’s largest, using data from the 2010 Census and school transfer requests in the 2019-2020 school year.

The Ecuadorian educational system is organized into three levels: Preschool (*Educación Inicial*), Primary School (*Educación General Básica*) and Secondary School (*Bachillerato*). In this paper, we focus on school assignments at the “entry level”—enrolment in Preschool 1, Preschool 2, and Primary 1.

Ecuador has three types of schools: public (*fiscal* and municipal), *fiscomisional* (partially state-funded), and private (fully family-funded). Nationwide, public schools account for 73.8% of enrolments, with *fiscomisional* at 6% and private schools at 20%. In Manta, public schools represent 66% of entry-level enrolments, while *fiscomisional* and private schools account for 4% and 30%, respectively.

This pilot focused on free public schools, so private schools were an outside option not included in our model, potentially leading us to overstate the welfare gains. A survey conducted after the application period is suggestive on this point, though it cannot settle it.³ Only 1.5% of the respondents who answered that item cited contingent private enrolment as their *principal* reason for not listing more alternatives, despite many applicants submitting just one or two preferences. The item asked for a single main reason and did not ask whether private schools entered a family’s choice set at all, so it establishes that few families named private options as the binding reason, not that private schools were absent from their consideration; the outside-option caveat therefore stands.⁴

2.1 Distance-Centric Algorithm

Before the COVID-19 pandemic, the school assignment system in Ecuador was based mainly on the applicant’s location, reported through the code on the family’s electricity account (CUEN). In addition to the linear distance criterion, a prioritization criterion determined the order in which applicants were processed (being processed first was preferable). This prioritization criterion was randomly assigned to students applying to be enrolled in the system.

The assignment system was part of a broader six-phase enrolment process, detailed in Appendix C. The Assignment Phase, the third of these phases, was conducted in stages. In the first stage, applicants were categorized based on their preference for regular, rural, bilingual, or special education programs. Where applicable, applicants with siblings already enrolled were assigned to the same school. Students applying for non-regular programs and those with

up the system in coastal districts beginning in 2023. Because of the COVID pandemic and the fact that the distance-centric alternative required several in-person interactions during the process, the Ministry is currently using a first-come, first-served digital system.

³The objectives of this survey were to gather information about parents’ overall satisfaction with the system, the information sources they used to apply for schools, and their awareness of the school supply, among other aspects. The survey drew 1,873 respondents, of whom 1,517 (81%) completed the full instrument.

⁴All applicants were ultimately assigned, though some were placed in a school outside their reported preferences. In these cases, the assigned school was the closest possible alternative, as explained in subsection 2.1.

siblings in the system were prioritized and processed before other applicants.

Once these groups were assigned through a process carried out directly at the district headquarters, the rest of the students were assigned using the distance-centric (DC) algorithm, which the Ministry called the “mathematical model.” Legal guardians could complete an individual registration (of a single applicant) or one for a “group of siblings.” While this latter option suggests that the system prioritized assigning groups of siblings to the same school over distance-based considerations, this was not confirmed by the Ministry experts with whom we interacted.

Regular assignments were processed as follows:

- At each level, random numbers were given to all applicants. These random numbers correspond to the prioritization criteria mentioned above and determined the processing order.
- Following this order, applicants were assigned to the closest school (linear distance) with vacancies, in an iterative process that used increasing distance radii from the applicant’s home.⁵

This procedure is an application of the “serial dictatorship” mechanism: applicants select schools one after the other. Given that the order of choice has a random component, such assignment models have been termed “random serial dictatorships” (Abdulkadiroğlu & Sönmez, 1998). Sibling-group enrolment thus appears to function as a priority, since the system tries to assign these groups to the same schools ahead of individual applications.

As explained above, an applicant’s home location was derived from the family’s electricity-account record. Like any assignment rule based on residential address, the distance mechanism is in principle manipulable through reported location, and an electricity record is an imperfect proxy for true residence (some households share a meter or lack one). We do not take a stand on how prevalent such misreporting was in Manta; we treat it instead as a robustness concern and bound its effect on our welfare comparison in Section 5.

The Ecuadorian government’s concern with minimizing the distance to school arises from public policy considerations, not from effects on other dimensions such as public expenditure on free busing to schools. The latter consideration is nevertheless relevant in other contexts (e.g., many US cities), meaning that analyses comparing assignment mechanisms in similar cases should consider inclusion of these budget factors.

2.2 Deferred Acceptance (DA) Mechanism

The pilot used the deferred acceptance mechanism (Gale & Shapley, 1962), following the best practices in school choice mechanism design (Pathak, 2011; Correa et al., 2019). The specification of the assignment algorithm included static and dynamic sibling priorities, family linking,

⁵The schools available were evaluated at radii of 100 m, 200 m, 300 m, and 500 m, and then at increments of 250 m up to 3.5 km.

and a multiple tie-breaking rule.⁶

The static and dynamic sibling priorities indicate that an applicant will be prioritized for assignment to a school/program if their sibling is already assigned to the school (static). If the applicant is applying at the same time with another sibling, and one of them is assigned to a school,⁷ the applicant that has not been assigned yet will receive priority for being assigned to that same school (dynamic). The dynamic sibling priority is lower than the static sibling priority because the latter is already defined (the sibling is attending the school), while the former will depend on the answer from the applicant after the assignment.

The family linking feature consists of trying to assign all siblings applying together to the same schools. Following a descending order, where older applicants are assigned first, if an older sibling is assigned to school A, the applications of the younger siblings will be modified to put school A as the first-ranked school to improve the probability of being assigned together. Finally, a multiple tie-breaking rule gives each applicant a different lottery number for each school to which they apply. Lottery numbers are used to break ties within priority groups when a school receives more applications than spaces available.

2.3 Mechanisms and Their Properties

We state the two mechanisms in market-design terms, because the comparison that follows is precisely one of how each treats applicants' *reported preferences* relative to their *location*. Both are *centralized, algorithmic* procedures: a single algorithm clears all schools simultaneously, before and after the reform, and both collect applications through the same digital platform (the prior distance process retained some in-person steps). What changes is thus neither the coordination of supply and demand nor the digitization of the process, but the data the algorithm acts on: DC allocates on location, DA on reported preferences.

Lemma 1 (The distance mechanism is a serial dictatorship over distance). *Fix the realized tie-breaking lottery (and the static-sibling pre-step). The distance-centric mechanism (DC) is the serial dictatorship in which applicants, ordered by the lottery within the static-sibling and regular priority classes, are processed in turn and each is assigned her nearest still-available program. Equivalently, DC is the applicant-proposing deferred-acceptance algorithm run on each applicant's own distance-sorted preference list—these differ across applicants, since distance rankings depend on home location—with a single tie-breaking lottery that induces a common applicant order at every school, and priorities {sibling $\prec$ regular}. Consequently, conditional on the sibling pre-step, DC is Pareto-efficient with respect to the remaining applicants' distance preferences; the pre-step itself can hold a seat for a sibling applicant that a nearer applicant would otherwise take, so the unconditional allocation is efficient only for sibling-adjusted distance preferences. In neither case does it condition on reported preferences.*

⁶The deferred acceptance algorithm was chosen for its stability and its ability to accommodate the pilot's priority structures (static and dynamic sibling priority, family-linked applications, and varied priority-quota combinations). Our counterfactuals run the engine in its strategy-proof configuration—static sibling priority and a single quota per program, with dynamic sibling priority and family linking disabled; the latter two introduce complementarities that can break strategy-proofness (Roth, 1984; Kojima et al., 2013; Nguyen et al., 2016). Section 2.3 states the resulting incentive and efficiency properties.

⁷This can happen if one sibling is older than the other and will depend on the order in which the algorithm is run. If it is descending, the older sibling will give dynamic priority to the younger sibling. If it is ascending, it will be the other way around.

In other words, DC is a random serial dictatorship that optimizes the *wrong* objective: it is efficient and (constrained) stable for distance, but it discards the preference information the choice system elicits.⁸ Deferred acceptance, by contrast, acts on the reported rank-ordered lists and gives reported programs strictly higher priority than the distance-imputed alternatives appended to each list. This difference in the data each mechanism uses translates into a sharp difference in incentives.

Proposition 1 (Strategy-proofness). *In the configuration we analyse—static sibling priority, a single quota per program, and school-specific tie-breaking—deferred acceptance is strategy-proof: submitting one’s true rank-ordered list is a dominant strategy for every applicant. This is not immediate from the textbook result, because here listing a program is itself what confers reported priority there, so priorities are not fixed independently of the applicant’s report. The dependence is, however, monotone in the applicant’s favour: adding a program to the list can only improve her priority at that program, and leaves her priority—and hence her prospects—at every program she ranks above it unchanged. No reordering or omission can therefore make her better off, and submitting the true ranking remains weakly dominant. The distance mechanism is strategy-proof in stated rankings (being a serial dictatorship) but does not use them, conditioning instead on reported residential location.*

We rely on the strategy-proofness of deferred acceptance when we treat the submitted lists as truthful in estimation (Section 5).⁹ Strategy-proofness disciplines the *order* in which a family ranks the schools it does list; it does not by itself imply that every unlisted school is less desirable than every listed one. Treating a short list as the truncation of a complete true ranking is a separate, empirical assumption of the demand model, and we return to what it does and does not deliver in Section 5. Because the distance rule conditions on residential location, it is—like any address-based assignment—in principle manipulable through reported location, a vulnerability deferred acceptance does not share. We do not rely on the magnitude of any such manipulation; we treat it only as a robustness bound (Section 5).

Remark 1 (Efficiency benchmarks). Following [Abdulkadiroğlu et al. \(2017\)](#), we benchmark DA against two efficiency refinements built from the DA outcome: stable improvement cycles (SIC, [Erdil & Ergin, 2008](#)) and top trading cycles (TTC, [Abdulkadiroğlu & Sönmez, 2003](#)). The chain $DA \rightarrow SIC \rightarrow TTC$ weakly Pareto-improves the DA assignment by construction; SIC is constrained-efficient within the set of stable matchings; TTC, run on the SIC allocation as an endowment, executes the remaining Pareto-improving trades. We use both as improvement operators on the realized assignment, not as standalone mechanisms: the standalone incentive properties of TTC do not transfer to this report-dependent composite, and the composite’s outcome need not be stable, which is why we read them as efficiency benchmarks rather than design candidates. In our environment the two benchmarks coincide (Proposition 2, Appendix E). The comparison between DA and DC, by contrast, is one of average realized welfare, not Pareto dominance: DA delivers higher mean utility because it assigns far more applicants

⁸The priority that rations seats under DC is the common lottery/processing order, not each applicant’s own distance to a school; DC is therefore not stable in the sense of prioritizing nearer applicants at a school (a closer applicant with a worse lottery draw can be displaced by a farther one with a better draw).

⁹Strategy-proofness of DA refers to the configuration we study (static sibling priority, single quota per program, multiple tie-breaking). The deployed platform additionally supports dynamic sibling priority and family linking, which, like couples in matching markets, can in principle break individual strategy-proofness. Our simulations disable both; Section 5 states how estimation handles the gap between this analysed configuration and the deployed one.

to a reported preference, even though individual applicants may be better off under either mechanism in a given draw.

3 Related Literature

This paper connects three strands of research on the design of school choice and assignment systems: the market-design literature that studies the theoretical properties of assignment mechanisms; the empirical literature that uses estimated demand to compare the welfare consequences of alternative designs; and the design literature on neighbourhood and distance-based priority that our distance-minimizing status quo instantiates. It also speaks to a growing body of work on the adoption of centralized choice and assignment platforms in developing countries. Our contribution is not a new estimator. We reuse the random-coefficients rank-ordered demand model of [Abdulkadiroğlu et al. \(2017\)](#) and [Rossi et al. \(1996\)](#) without modification, precisely so that our welfare estimates remain comparable to that literature; what we change is the counterfactual it is asked to evaluate, and the data that go into it. The counterfactual pair is a coordinated *distance-minimizing* status quo against a coordinated *preference-respecting* alternative, and the comparison carries external validity to a low-income, early-grade Latin American setting. The data are the two inputs this literature usually takes as given. The first is travel cost: the near-universal specification, linear in straight-line distance, is misspecified, and a concave function of *walking* time predicts the submitted rankings markedly better (Table 4) — with the consequence, developed in Section 5.3, that dispersion in the distance coefficient is better read as an upper bound on taste heterogeneity than as a measure of it. The second is the schools themselves, whose observable characteristics are thin in administrative data and which we measure from satellite and street-level imagery.

The modern analysis of school assignment begins with the application of two-sided matching theory ([Gale & Shapley, 1962](#)) to school choice as a mechanism-design problem ([Abdulkadiroğlu & Sönmez, 2003](#)). A large theoretical literature characterizes the trade-offs among the leading mechanisms—the student-proposing deferred acceptance (DA) algorithm, the Boston (immediate-acceptance) mechanism, and top trading cycles—along the dimensions of stability, ex-post efficiency, and strategy-proofness ([Abdulkadiroğlu & Sönmez, 2003](#); [Abdulkadiroğlu et al., 2005](#); [Pathak, 2011, 2017](#)), building on the house-allocation and random serial dictatorship literature ([Abdulkadiroğlu & Sönmez, 1998](#)). Because DA can be inefficient when priorities involve ties, subsequent work studies tie-breaking and Pareto-improving refinements such as stable improvement cycles ([Erdil & Ergin, 2008](#); [Abdulkadiroğlu et al., 2009](#); [Ashlagi et al., 2019](#)), while a parallel literature examines the manipulability of the Boston mechanism and the distinction between sincere and sophisticated play ([Pathak & Sönmez, 2008](#); [Ergin & Sönmez, 2006](#); [Abdulkadiroğlu et al., 2006, 2011](#); [Chen & Sönmez, 2006](#); [Featherstone & Niederle, 2016](#)). A recent large-market literature clarifies the terms of the efficiency–stability trade-off that underlies our welfare benchmarks: [Che & Tercieux \(2019\)](#) show that the efficiency cost of insisting on stability can be substantial in large markets, while [Abdulkadiroğlu et al. \(2020a\)](#) characterize empirically the joint trade-off among efficiency, justified envy, and strategy-proofness across serial dictatorship, DA, and top trading cycles—the very mechanisms our comparison spans, since the distance status quo is itself a serial dictatorship over distance (Lemma 1). We draw on this literature both to motivate the choice of DA for the Manta pilot and to define the set of

counterfactual mechanisms—DA, stable improvement cycles, and top trading cycles—against which we benchmark welfare.

Our central contribution belongs to the empirical literature that estimates student demand in order to quantify the welfare effects of mechanism choice; [Agarwal & Somaini \(2020\)](#) survey the revealed-preference methods this literature uses to recover preferences from the rank-ordered lists generated by potentially manipulable mechanisms. The closest antecedent is [Abdulkadiroğlu et al. \(2017\)](#), who study New York City’s move from an uncoordinated, multiple-offer system to a coordinated, deferred-acceptance mechanism and find that the welfare gains from coordination per se dominate those from any further algorithmic refinement, with the largest gains accruing to students who would otherwise have been administratively assigned. [Agarwal & Somaini \(2018\)](#) develop methods to estimate preferences from manipulable mechanisms and compare allocations across designs, and [Calsamiglia et al. \(2020\)](#) structurally estimate a model of school choice to compare the Boston mechanism with its alternatives. A theoretical counterpoint cautions that an ordinal, strategy-proof mechanism need not raise cardinal welfare relative to a manipulable one ([Abdulkadiroğlu et al., 2015](#)), underscoring the need for an explicit cardinal demand model in any welfare comparison. None of these papers, however, provides a comparison in which coordination is held fixed and only the *preference content* of the mechanism varies: each contrasts a coordinated design with an uncoordinated or differently coordinated one. The nearest thing to our comparison in that literature is the “neighbourhood assignment” benchmark that [Abdulkadiroğlu et al. \(2017\)](#) themselves construct as a hypothetical lower bound, together with the realized walk-zone and neighbourhood-priority designs studied by [Dur et al. \(2018\)](#) and [Laverde \(2024\)](#)—but here that benchmark is not hypothetical: it reproduces a rule that was in force in the same market immediately before the reform, and that remains the operative counterfactual for the many systems that still assign by proximity. We do not claim an observational advantage: as in [Abdulkadiroğlu et al. \(2017\)](#), no applicant’s outcome under the benchmark rule is observed for this cohort, and our distance arm is simulated. What differs is that the rule being simulated is a policy that actually governed this market until immediately before the reform, not a hypothetical construct. Our setting therefore supplies exactly the fixed-coordination comparison this literature lacks: a single centralized algorithm cleared the Manta market both before and after the reform, so our exercise holds the coordination of supply and demand fixed and changes only the objective the algorithm optimizes—from assigning each applicant to the closest available school to respecting truthfully elicited preferences. Ours therefore complements New York rather than repeating it: that transition bundles coordination together with the move to a preference-respecting algorithm, whereas ours isolates the component the bundle cannot separate—and does so in a small coastal town, not a large, wealthy city.

A methodological literature studies how to recover preferences from data generated by assignment mechanisms. Estimation is most transparent under strategy-proof mechanisms, where truthful reporting is a dominant strategy; we follow [Abdulkadiroğlu et al. \(2017\)](#) and a Bayesian rank-ordered approach ([Rossi et al., 1996](#)) in treating submitted lists as truthful and estimating a random-utility model by Gibbs sampling. Recent work qualifies this assumption: [Fack et al. \(2019\)](#) argue that data from deferred acceptance robustly identify stable matchings, not literal truth-telling and propose estimators based on weaker assumptions, and [Artemov et al. \(2023\)](#) show that reported lists may contain payoff-irrelevant “mistakes” when application is costly—concerns that bear directly on the substantial share of applicants in our data

who submit short lists. Two features of our setting limit these concerns—applicants faced no cap on the length of their ranked lists, and the pilot’s design and communication emphasized strategy-proofness, which holds exactly for the static-priority, single-quota configuration we analyse (Section 2.2)—but we also assess robustness to behavioural departures, including strategic misreporting of residential location. A distinct point concerns the object our demand model recovers: because we denominate utility in distance and include no measure of school quality, the estimated valuations are realized, revealed valuations of schools, not parents’ assessments of causal effectiveness. [Abdulkadiroğlu et al. \(2020b\)](#) show that families’ rankings reward schools attended by high-achieving peers more than schools with high causal value-added; our welfare object is deliberately agnostic on this distinction, measuring how families trade off their realized valuation of a school against distance, whatever the source of that valuation. Our structural approach is shared by a broader set of demand-based studies of school choice and admissions ([Kapor et al., 2020](#); [Idoux, 2022](#); [Calsamiglia et al., 2020](#)); a complementary reduced-form literature documents how information and the local supply of schools shape enrolment without recovering demand primitives ([Hastings & Weinstein, 2008](#); [Hastings et al., 2016](#); [Gilraine et al., 2021](#)).

Because the reform we study replaces a distance-based rule with a preference-based one, our results bear directly on the market-design literature on neighbourhood and distance-as-priority assignment, of which the distance-minimizing status quo is a limiting case. [Dur et al. \(2018\)](#) show, in the context of Boston’s walk-zone reserves, how distance-based priority interacts with the rest of a mechanism in ways that are often opaque to the families it is meant to serve; the Manta status quo is the extreme of this design, a system in which the menu an applicant can receive is ordered by proximity alone and preferences are not solicited at all. The mapping is close but not exact: under DC distance orders each applicant’s *options*, while priority itself comes from the sibling classes and the common lottery (Lemma 1). [Calsamiglia et al. \(2021\)](#), in this journal, study how neighbourhood priority and the transportation costs it economizes on interact with risk aversion to shape cardinal segregation, and [Laverde \(2024\)](#) documents how the distance families must travel under choice systems shapes equal access relative to a neighbourhood-assignment benchmark. These papers also discipline our mapping of the distance status quo to a neighbourhood benchmark—it is closely analogous to, but not identical to, a single-zoned-school assignment, since the capacity-constrained, nearest-available rule leaves a non-trivial share of children placed at a school other than their nearest one.

Finally, our paper contributes to a rapidly growing literature on centralized choice and assignment in developing countries, where many systems are transitioning from decentralized or distance-based allocation to coordinated digital platforms ([Elacqua et al., 2016, 2021](#); [World Bank, 2024](#)). Work in these settings has emphasized the role of information provision and platform design in shaping choices and equilibrium outcomes ([Allende et al., 2019](#); [Arteaga et al., 2022](#); [Kapor et al., 2020](#)), with field evidence from Ghana, where a centralized choice system and information alone proved insufficient to close gaps in placement ([Ajayi et al., 2026](#); [Ajayi & Sidibé, 2020](#)), Pakistan ([Andrabi et al., 2017](#)), and the United States ([Hastings & Weinstein, 2008](#); [Hyman, 2020](#)), and laboratory evidence on how the mechanism itself shapes information acquisition ([Chen & He, 2021](#)). Related work on the region’s centralized systems studies the consequences of the admission decisions they produce, not the information families bring to them ([Fabregas, 2018](#)). The Chilean experience is especially relevant: a national move to a deferred-acceptance system ([Correa et al., 2019](#); [Larroucau & Ríos, 2021](#)) has been accompanied

by evidence on application mistakes and information frictions that generate short rank-order lists (Larroucau et al., 2025)—a margin we confront directly—and by evidence that centralized admission can shift, rather than eliminate, school segregation (Kutscher et al., 2023). Despite widespread adoption, direct estimates of the welfare effects of moving to a preference-based coordinated system remain scarce outside high-income contexts, the New York City evidence of Abdulkadiroğlu et al. (2017) being the leading exception. The closest developing-country antecedent is Arcidiacono et al. (2021), who estimate the welfare value of school choice from a two-stage voucher experiment in rural India and—unusually—pre-commit the structural model and its predictions before examining the experimental data, a design that lends external credibility to demand-based welfare evaluation of school choice in low-income settings. Their gains, however, arise from *expanding* families’ choice sets to include private schools; ours hold the choice set and market coordination fixed and isolate the value of an algorithm acting on the preferences families already hold. We provide such estimates for a Latin American city, situating the study within a broader regional effort to improve the efficiency and equity of education systems—from school finance and technical education (Elacqua et al., 2019b,a) to teacher allocation (Ajzenman et al., 2024) and the expansion of instructional time (Bellei, 2009; Hincapié, 2016; Valenzuela, 2005; Berthelon et al., 2016; Cerdan-Infantes & Vermeersch, 2007).

Because the reform also alters the sorting of families across schools, our results connect to the literature linking school choice to residential location and neighbourhoods: choice can change the sorting of students across schools and the capitalization of school quality into housing prices (Epple & Romano, 1998; Urquiola, 2005; Kane et al., 2006), and systems that disperse neighbourhood cohorts across many schools raise transportation costs and may weaken community ties (Ashlagi & Shi, 2014; Ebbert & Ullman, 2011; Sutherland, 2012); Calsamiglia et al. (2021) show how the neighbourhood priority that economizes on these costs can itself reshape segregation. Our welfare analysis quantifies the central trade-off these concerns embody: a preference-respecting mechanism assigns some families to schools farther from home, and because our utility estimates are denominated in distance with the distance coefficient normalized to -1 , the welfare gains we report are already net of this additional travel—they measure valuations of preferred schools *after* families have paid the price of additional proximity foregone.

4 Data

The data used in this paper come from the centralized choice and assignment system (CCAS) pilot web page created in 2021 in the region of Manta, Ecuador.¹⁰ The first data set comprises the supply of vacancies for all schools and programs offered in the pilot, where an educational program consists of a combination of grade and school. The pilot was implemented for all students entering Preschool 1, Preschool 2, and the first year of primary school (i.e., ages 3 to 5) for the first time. Vacancies are presented in Panel A of Table 1.

Preschool 1 has the most vacancies. Aggregate seat counts, however, are a poor guide to how binding capacity is, because seats are not located where applicants are: Montecristi alone offers 654 Primary 1 places against 124 local applicants, so the grade with the most seats per applicant is also the one in which placement is hardest. What matters for the comparison

¹⁰All personally identifiable information was removed from the data before they were shared with us.

that follows is whether an applicant can be placed in a school she would actually list, and by that measure congestion rises steeply across the entry grades—deferred acceptance places 96% of Preschool 1 applicants in a listed school, 71% in Preschool 2 and only 47% in Primary 1 (Tables 11–13). It is in this sense that we call Preschool 1 the least and Primary 1 the most congested grade throughout: an equilibrium measure of effective placement difficulty, which bundles seats-where-applicants-are with list length and preference concentration, not a pure capacity primitive.¹¹

The second data set contains information about students and their legal guardians, including geolocation, sibling relationships, special educational needs, and nationality.¹² Each applicant’s rank-ordered list (ROL) of preferences had no length limit, and lotteries were conducted for each program. As noted, applications were supplemented by adding all other alternatives, sorted by distance, to the initial preference list in case the applicant was not assigned to a preferred school. In the pilot, each program listed by the student had a different lottery number, while a single number was used across the appended list of alternatives. The simulations reported below draw the lottery independently for every program, appended ones included; the consequence of that departure for the efficiency refinements is taken up in Section 6.

Because both applicants and schools are geolocated, that spatial mismatch can be seen directly, and it is sharpened by how uneven Manta is within its urban core. Figure 1 shades census blocks by neighbourhood education and overlays the public schools in the assignment system: adjacent neighbourhoods differ sharply in schooling, so the local menu a distance rule imposes on a family is drawn from starkly different surroundings over short distances. The map documents that spatial unevenness; whether it translates into differently *valued* nearest schools is what the estimated school effects and the rankings in Figure 2 speak to, and the welfare incidence we estimate below is where the two meet.

The distribution of applicants by geographic unit and grade is presented in Panel B of Table 1. Notably, at least in the case of the geographic unit of Manta (Cantón), the numbers of applicants in Preschool 1 and Preschool 2 are roughly equal. Although this poses a public-policy challenge, since it is desirable to enrol students earlier, it is also an interesting dynamic for the application system: families that postpone enrolment put themselves at a strategic disadvantage, because there are fewer seats in Preschool 2 once currently enrolled Preschool 1 students move automatically to the next level.

Figure 7 of Appendix A provides an overview of applicant priorities and the lengths of rank-ordered lists. Note that most applicants declared only a single preference despite there being no limits placed on the length of the preference list. This may be a legacy of the previous system, in which applicants did not choose a portfolio of schools and were largely assigned to a school based on linear distance rather than their preferences.

A complementary explanation for the large number of short application lists is that, as shown in Figure 8 of Appendix A, applicants surveyed after the application period indicated

¹¹This is likely due to several factors: i) students prefer schools closer to home, and schools in more crowded areas were filled under the previous distance-based algorithm; ii) students dissatisfied with their assigned school can request a transfer.

¹²The preferences of applicants with siblings already enrolled at their school of interest are detailed in Panel A of Table 20 in Appendix B. We do not have data on applicants whose siblings are enrolled in schools not listed in their preferences.

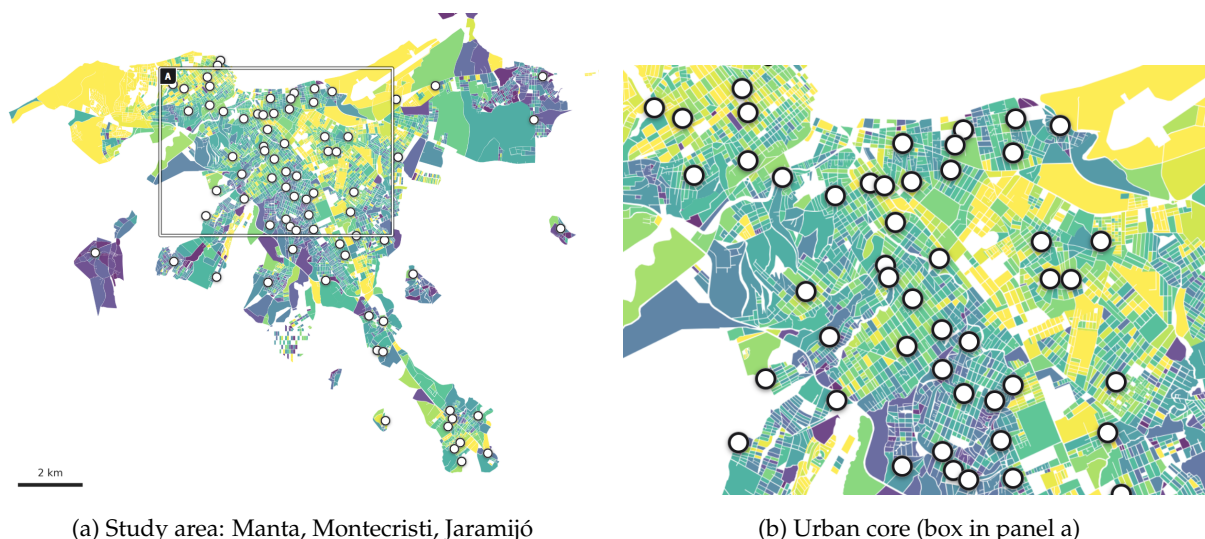


Figure 1: Neighbourhood education and the location of schools in Manta. Census blocks (Censo 2022) are shaded by a measure of neighbourhood education (average years of schooling among adults); white markers are the public schools in the assignment system. Panel (a) shows the full study area with the dense urban core boxed; panel (b) zooms to that core, where block-to-block differences in education are pronounced. Adjacent neighbourhoods differ sharply in schooling, so the default a distance rule imposes—whatever nearest school a family happens to have—is drawn from very unevenly situated local menus. The figure documents that spatial unevenness; it does not by itself establish that families value those nearest schools differently, which is what the estimated school effects and the rankings in Figure 2 speak to.

Note: Census blocks are from the 2022 Ecuadorian Census (INEC), shaded by the average years of schooling of resident adults; white markers are the public schools in the assignment system. Source: authors’ calculations.

high expectations of being assigned to their top choice. These responses were collected before results were published to avoid bias. When asked why they did not add more programs to their rank-ordered list (ROL), of the 1,639 respondents (87.5%) who answered this item, 55.0% reported that no public school was sufficiently close, 33.1% were confident they would be assigned to their top choice, 6.3% found it difficult to find more schools, 4.1% preferred no assignment over adding more options, and 1.5% preferred a non-public school (outside option).¹³ The dominant reason is thus the scarcity of acceptable nearby options, not an inability to locate them—only 6.3% cite difficulty finding more schools. Families’ knowledge of the alternatives they did *not* apply to is nonetheless limited: shown up to five nearby public schools outside their application, respondents report knowing 7.7% of them well and a further 21.6% by name only. Families themselves thus attribute short lists to scarcity of acceptable nearby options or to confidence of placement rather than to difficulty finding schools—though the item takes a single principal reason, and with most unlisted nearby schools unknown, awareness

¹³The verbatim option is “No hay más instituciones educativas fiscales lo suficientemente cerca”—there are no further public schools sufficiently close. An earlier draft of this paper glossed this category as families lacking information on nearby alternatives; the option text does not support that reading, and we report it here as the scarcity response it is. The difference between the 55.0% reported here and the 56% of that earlier draft is one of denominator—all respondents who answered the item, rather than those who completed the full instrument—not a revision of the data.

Table 1: Vacancies and Applicants by Geographic Unit (*Cantón*) and Grade

Panel A: Vacancies			
Cantón	Preschool 1	Preschool 2	Primary 1
Manta	1,830	1,394	425
Montecristi	905	668	654
Jaramijó	110	47	37
Total Grade	2,845	2,109	1,116
Total Global	6,070		

Panel B: Applicants			
Cantón	Preschool 1	Preschool 2	Primary 1
Manta	1,101	1,143	338
Montecristi	481	437	124
Jaramijó	125	125	107
Other	2	0	1
Total Grade	1,709	1,705	570
Total Global	3,984		

Note: Vacancies and applicants by canton and grade in the 2021 Manta pilot. Source: authors' tabulation of Ministry of Education application records.

frictions may also have contributed—and short lists do not represent a fully informed comparison among alternatives; Section 5 discusses how the demand model treats single-school lists and Section 6.4 the consequences for the welfare comparison.

In any case, because the CCAS was new to families in Manta, they likely did not fully adapt their behaviour to the new rules, and so may not have taken full advantage of parental choice and preference reporting. How this affects the comparison is ambiguous. Familiarity could lengthen lists and improve deferred-acceptance placements, raising the measured gain; but families who discover acceptable schools they had not known of would also raise the modelled value of the distance rule's assignments, and participation and congestion could change with them. Without a later cohort we cannot sign the net effect relative to the long run.

Estimation sample. We estimate preferences for the applicants whose ranked schools all lie within the in-market set of public schools we model. Of the 3,984 applicants (Table 1), this retains 2,372; the remainder listed at least one school outside that set—out-of-region, private or *fiscomisional*, or night-only and de-duplicated programs—or only the distance-imputed fallbacks, and so fall outside the demand model (Table 2). The welfare comparison (Table 8) is computed over the 2,200 of these applicants who receive a comparable placement in at least one lottery draw: 1,019, 838 and 343 by grade (Table 2). A draw contributes to an applicant's average only if she is placed under *all four* mechanisms in that draw, so the draws being compared are common across mechanisms by construction, not mechanism-specific; the simulation records a draw for an applicant with all four utilities or not at all. Almost every retained applicant qualifies in all 100 draws. The exception is Primary 1, where 16 qualify in 86 to 99 and are averaged over those, so their means rest on slightly fewer draws than the rest. The 172

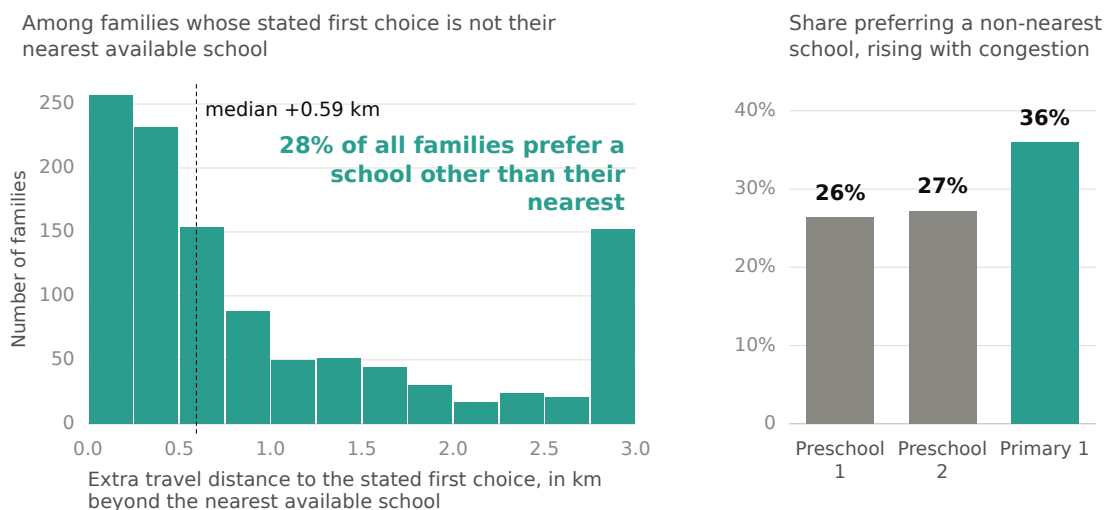


Figure 2: Families reveal preferences for schools other than their closest. Left: among the 28% of families whose stated first choice is not their nearest available school, the additional travel distance that first choice entails (median +0.59 km). Right: the share of families preferring a non-nearest school, by grade, rising with congestion. A distance-minimizing rule treats these preferences as costs; a preference-based mechanism acts on them. This dispersion of preferences away from proximity is the source of the welfare gains we estimate.

Note: All applicants in the three grades. The nearest school is the nearest offering the applicant’s grade, by straight-line distance; a first choice counts as non-nearest only if it lies more than 100 metres farther than that school. Source: authors’ calculations from the 2021 Manta application data.

excluded applicants (79, 47 and 46 by grade) are absent from the simulation files the welfare comparison reads, which cover 3,698 of the 3,975 applicants the assignment simulations place; they are not missing at random, since the shortfall in those files is proportionally largest in the most congested grade—11.8% of Primary 1 applicants, against 7.2% and 5.3% in Preschool 1 and 2—and the applicants those files omit place worse under either rule. The welfare sample is therefore selected toward the applicants the market can accommodate, whereas the assignment shares (Table 7) describe the full applicant pool the pilot actually assigned; the two therefore have different denominators, and we read the access gain as a statement about all applicants and the welfare gain as one about the applicants whose preferences we can estimate *and* who receive a comparable placement in the simulations.

Table 2: From applicants to the estimation sample

	Preschool 1	Preschool 2	Primary 1	Total
Applicants	1,709	1,705	570	3,984
Preferences estimated (in-market choice set)	1,098	885	389	2,372
Welfare-comparison sample	1,019	838	343	2,200

Note: “Preferences estimated” keeps applicants whose ranked schools all lie within the modelled in-market public-school set; the remainder listed at least one out-of-market, private, *fiscomisional*, night-only, or de-duplicated school, or only distance-imputed fallbacks. The welfare-comparison sample is the subset of these for whom the per-student welfare distribution is computed across the 100 simulated lottery draws.

5 Estimating Preferences and Comparing Mechanisms

Our analysis in this section treats applicants’ reported preference orderings as accurate representations of true family preferences. That is a maintained assumption, and this is the place to be precise about what supports it. Truthful reporting is weakly dominant in the static-priority configuration we analyse (Proposition 1), and the platform’s communication emphasized exactly that; but the deployed mechanism also offered dynamic sibling priority and family linking, which can break individual strategy-proofness for the applicants they touch, so the incentive result covers the analysed configuration rather than the deployed one exactly. Two facts bound the residual gap: the affected group—applicants with a sibling applying in the same round—is a small minority, and dropping the sibling term together with the sibling-linked applicants leaves the estimates and welfare conclusions unchanged (Appendix B.1).¹⁴ We thus compare the share assigned to a preferred option under each alternative and use reported preferences to estimate welfare differences. Our welfare analysis is structural and, by design, model-based: because each family experiences only one mechanism, the equilibrium welfare effect of an alternative rule cannot be observed directly but must be recovered from estimated preferences and simulated under the counterfactual—the standard approach to policy questions that cannot be randomized. We follow a substantial literature that studies education markets this way, recovering demand primitives from administrative micro-data and simulating policy counterfactuals (Neilson, 2026; Kapor et al., 2020; Abdulkadiroğlu et al., 2017; Idoux, 2022; Calsamiglia et al., 2020); in its terms, we estimate the demand block of a centralized, non-price education market (Agarwal & Somaini, 2020), holding the supply of schools fixed.

To estimate preferences, we adopt the utility model of Abdulkadiroğlu et al. (2017) and apply their Markov Chain Monte Carlo (MCMC) estimation method using Gibbs sampling (Rossi et al., 1996), similar to recent applications in the school choice context (Kapor et al., 2020; Idoux, 2022). Our methodology encounters two key challenges: a limited set of school covariates and the assumption that applicants lack full knowledge of all available schools. To address the first, we estimate a model without covariates, where differences in school appeal are driven by a school-specific effect, unobservable to the econometrician but assumed to be known to families. We also introduce a random coefficient for the distance parameter to capture variations in how families value a school’s utility relative to distance. For the second challenge, we work with a geographically constrained set of alternatives, treating every school in the applicant’s region as available to be ranked. This is a modelling convenience, not a description of behaviour: our companion survey indicates that families know only a minority of the nearby public schools they did not apply to (Section 4). We retain it because the alternative—estimating consideration sets jointly with preferences—is beyond our scope, and Section 6.4 sets out what it implies for the welfare comparison and in which direction.

We compare the distance-centric (DC) algorithm described in subsection 2.1 with the Deferred Acceptance (DA) algorithm, using the Stable Improvement Cycles (SIC) (Erdil & Ergin, 2008) and Top Trading Cycles (TTC) (Abdulkadiroğlu & Sönmez, 2003) algorithms as benchmarks. The TTC step serves as our welfare benchmark: run on the SIC allocation as an en-

¹⁴We probe this assumption directly in Appendix B.1: re-estimating demand by the stability approach of Fack et al. (2019)—which assumes only that the realized matching is stable, so each applicant weakly prefers her assignment to every school she could ex post have been admitted to, and hence does not require the reported list to be truthful—leaves the entry-grade welfare conclusion unchanged.

dowment, it executes any remaining Pareto-improving trades, so the chain weakly dominates SIC—which produces a stable but welfare-constrained assignment—and likewise weakly dominates DA. The ordering is weak by construction, and in our environment it is not strict: SIC and TTC coincide in every simulation (Proposition 2), and their improvement over DA is positive but very small in the entry grade. To ensure a consistent welfare comparison, we follow the approach in Abdulkadiroğlu et al. (2017), first running the SIC algorithm over each DA assignment, and then applying TTC to the resulting DA-SIC assignment.¹⁵ We conduct 100 lottery simulations for both the DA and DC algorithms to derive our welfare calculations, as in Abdulkadiroğlu et al. (2017).

To compare mechanisms, we first re-run the DA algorithm used in the pilot. We use the same inputs but exclude students with special needs, to make the assignment comparable to that of the DC algorithm.¹⁶ In the implemented DA, the reported preference rankings were appended to all non-ranked programs using a linear distance sorting criterion. Applicants received a lower priority in the distance-imputed preferences to maximize assignment to the reported preferences.¹⁷ We define assignments to imputed preference as non-preference assignments to distinguish them from the overall assignment obtained with the DC algorithm.

To replicate Ecuador’s previous system (described in subsection 2.1), we consider all available programs and rank them using linear distance sorting. Students with siblings in the system were assigned (if possible) to their sibling’s school before the main process was initiated. To this end, we create a priority group for these students that only applies at the schools in which their siblings are enrolled. This priority is followed by a priority for groups of siblings applying together, as these groups were processed before individual applicants in the main process; this priority is thus applied to all available programs. The Ministry did not confirm how joint sibling applications were prioritized (Section 2.1), so this is a reconstruction rather than a documented rule. Because changing the processing order in a serial dictatorship can move assignments beyond the families directly affected, we quantify the sensitivity directly. Re-running the distance mechanism on the same 100 lottery draws under two alternative rules that bracket the reconstruction—processing all joint-sibling groups before every individual applicant, and stripping sibling priority altogether—moves the share assigned to a listed preference by at most 1.0 percentage point, leaves the entry-grade welfare comparison unchanged (the recovered share stays at 99.6%), and moves Preschool 2’s recovered share by +1.5 to +4.5 points; only imprecisely-identified Primary 1 is materially sensitive, consistent with the wide interval we report there. Both alternatives *worsen* the distance rule’s assignment, so the reconstruction we use is the most favourable to the status quo among the three, and the welfare gains we report are on this margin conservative. Finally, given that applicants were processed sequentially, we run a single tie-breaking lottery to break ties.

¹⁵In this case, SIC and TTC produce the same assignment, meaning no Pareto efficiency improvements are attainable by relaxing stability constraints.

¹⁶We eliminate both students and vacancies related to special needs, which account for only 0.23% of applicants. This decision was made because students with special needs had a special assignment round before the regular one.

¹⁷When referring to applicant preferences, we intend reported preferences without the distance-imputed preferences.

5.1 A Random-Utility Model of School Choice

To estimate welfare, we first need to estimate the parameters determining the utility that families would receive from an assignment to a particular school. To this end, equation 1 presents our utility model.¹⁸

$$\begin{aligned} u_{ij} &= S_{ij}\lambda + \delta_j + (-1 + \gamma_i)d_{ij} + \epsilon_{ij} \\ \delta_j &= \bar{\delta} + \xi_j \\ \bar{\delta} &\equiv 0 \end{aligned} \tag{1}$$

$$\gamma_i \sim \mathcal{N}(0, \sigma_\gamma^2)$$

$$\xi_j \sim \mathcal{N}(0, \sigma_\xi^2)$$

$$\epsilon_{i,j} \sim \mathcal{N}(0, \sigma_\epsilon^2)$$

Here, S_{ij} is a dummy variable equal to 1 if applicant i has a sibling in school j , d_{ij} is the distance between applicant i and school j , and ξ_j is a school-specific preference that is unobservable to the econometrician but that families do observe when comparing alternatives. To identify the parameters we need both a location and a scale normalization for utility. We fix the location by setting $\bar{\delta} = 0$, as in [Abdulkadiroğlu et al. \(2017\)](#), and the scale by normalizing the average disutility of a kilometre of distance to -1 ; it is the latter that denominates utility in kilometres. We also include random coefficients over distance to school, γ_i , to capture heterogeneity across applicants in the relative importance of school attributes and travel distance.¹⁹ Because this specification expresses utility in distance units (km) – a result of imposing a -1 parameter on the average dis-utility of linear distance to school – a random coefficient on distance is much like a random coefficient on school attributes as in [Abdulkadiroğlu et al. \(2017\)](#): both shift the relative importance of distance and school attributes.

The specification in equation 1 assumes that an applicant’s utility increases when a sibling is already enrolled in a school. It does not, however, account for the possibility that the sibling enrolled there because the family liked the school in the first place. This implies that ϵ_{ij} is not

¹⁸Here, we adapt the model used by [Abdulkadiroğlu et al. \(2017\)](#), estimated through Gibbs sampling ([Rossi et al., 1996](#)). We estimate utility by MCMC, using the same conjugate priors, specifically the Inverse-Wishart distribution. The full utility specification, including priors, is provided in Appendix D.

¹⁹Conceptually, in each iteration of the Gibbs sampler, utilities are drawn using the estimated parameters of the previous iteration, using reported preference rankings to restrict possible values. Specifically, assuming that i ’s ranking is of size R (1 being the most preferred alternative and R the least preferred), utilities are drawn iteratively using a truncated normal distribution so that:

$$u_{i,j(r=1)} > u_{i,j(r=2)} > \dots > u_{i,j(r=R)} > u_{i,j(r=\hat{r})}, \forall \hat{r} > R$$

The sampler imposes this from both directions. A school ranked r is drawn between the current utility of the school ranked $r + 1$ below it and that of the school ranked $r - 1$ above it, the top-ranked school having no upper bound; the *last* ranked school is instead bounded below by the running maximum utility among that applicant’s unranked schools, and the unranked schools are then drawn below the last ranked one. When $R = 1$ the sole listed school is simultaneously first and last, so it is drawn from $(\max_{\hat{r} > R} u_{i,j(\hat{r})}, \infty)$ —unbounded above, but bounded below by the best unranked alternative—which is what enforces the final inequality above for single-list applicants.

random for such cases, highlighting the bias in the estimation. In Appendix B.1 we therefore present our preference parameter estimates and welfare calculations with no sibling-related considerations.

As in Abdulkadiroğlu et al. (2017), identification relies on the assumption that families report their preferences truthfully and consider all the alternatives within their geographic unit. Likewise, the key conditional independence assumption is that

$$(\gamma_i, \epsilon_{ij}) \perp d_{ij} | \xi_j$$

which, in our case, implies that conditional on the vertical school-specific parameter, unobserved tastes for programs are independent of linear distance to school. Two features of our setting make this assumption credible. First, for the applicants who rank two or more schools—between 33 and 43% of the sample, depending on grade (Section 4)—the rank-ordered lists provide *within*-applicant variation in distance across the schools a family ranks, so the trade-off between distance and the school effect ξ_j is identified from how each family orders its own listed schools, not from the cross-family location of homes. For applicants who list a single school, identification comes instead from the truncation that the listed school is preferred to every unranked alternative—including any that are nearer—together with the cross-sectional distribution of choices and the parametric structure; reassuringly, the welfare conclusions are unchanged when we drop the random coefficient or the sibling term (Tables 18 and 19 of Appendix B.1), which bounds the influence of that structure. The restriction as displayed applies to $(\gamma_i, \epsilon_{ij})$ jointly, and within-applicant variation in distance does not on its own dispense with the requirement on the random coefficient γ_i ; we therefore maintain it for both. Appendix B.1 reports that the welfare conclusions are unchanged when γ_i is dropped altogether, which bounds how much the estimates can be leaning on it. Second, we study entry-grade applicants (preschool and early primary), for whom residence is largely *predetermined* relative to this one-time application: families do not relocate their home in response to the menu of schools their three- to five-year-old can attend. The usual threat to the assumption—residential sorting, whereby families with a strong unobserved taste for a school move close to it (Urquiola, 2005; Kane et al., 2006)—is therefore muted, and true distance is plausibly independent of unobserved taste conditional on ξ_j .

Credibility of the truthful-reporting assumption. Because the configuration we analyse uses applicant-proposing deferred acceptance with static priorities and a single quota per program, submitting one’s true ranking is a weakly dominant strategy, a property the pilot’s communication strategy emphasized. (The deployed system additionally offered dynamic sibling priority and family linking, which relax this guarantee; we disable both in our analysis, as detailed in Section 2.2.) Nonetheless, recent work shows that data from deferred acceptance robustly identify *stable* matchings rather than literal truth-telling, and that stated rankings may depart from true preferences when submitting a longer list is costly or when applicants hold coarse information about their admissions chances (Fack et al., 2019; Artemov et al., 2023). Two features of our setting limit this concern. First, applicants in Manta faced no cap on the length of their ranked lists, which removes the list-truncation margin responsible for most documented departures from truth-telling in systems with short lists, such as the twelve-choice limit studied in Abdulkadiroğlu et al. (2017). Second, applications were submitted through an assisted platform that presented the feasible alternatives within the applicant’s geographic unit, mitigating

the coarse-information channel; the platform therefore displayed feasible alternatives to every applicant, whether or not the family had previously known them.²⁰ Moreover, in the strategy-proof configuration we analyse the two leading identifying assumptions in this literature point the same way, though they are not equivalent: under weak dominance a truthful list is optimal, while the stability-based estimator of Fack et al. (2019) sets the reported ranking aside altogether and requires only that each applicant weakly prefer her assignment to the schools she could have been admitted to. Its inequalities are strictly weaker, and it uses the realized assignment and ex-post feasible sets, not the submitted list; that both deliver the same welfare conclusions tells us the estimates do not hinge on the literal ordering of the lists families submitted. It does not validate the separate assumption that omitted schools were known and less preferred—both estimators retain preference inequalities over omitted feasible schools—which is why awareness is treated on its own terms in Section 6.4. Finally, a distinct concern is that behaviour under the counterfactual distance-based system—specifically, strategic misreporting of home location—could narrow the gap we attribute to deferred acceptance. We address this directly with a counterfactual that lets a large share of applicants misreport their address; even so, the listed-preference share stays well short of deferred acceptance’s, because a falsified address can signal a preference for at most one alternative (Table 21, Appendix B).

Short lists and the random coefficient. Most applicants submit short lists—about 57 to 67% rank a single school across the three grades, and only 12 to 21% rank three or more—so it is worth being explicit about what identifies the welfare gain for the short-list majority. The per-family random coefficient on distance, γ_i , is estimated only for applicants who rank more than two schools; for everyone else it is held at the normalized population value, so a single-list family carries *no* family-specific taste parameter and enters the estimates only through the cross-sectionally identified school effects ζ_j and the requirement that its listed school be preferred to every unranked alternative, including nearer ones. Three checks confirm that the headline does not rest on the structure imposed on short-list families. First, shutting the random coefficient off entirely—for *all* applicants—reproduces the welfare almost exactly: deferred acceptance still recovers 99.6% and 84.4% of the attainable range in the two preschool grades, against 99.6% and 82.3% in the main specification (computed as $1 - \Delta_{DA}/\Delta_{DC}$ from Panel A of Table 18 of Appendix B.1, which reports the mean-utility levels, not the share). Second, the gain is spread across list lengths, not concentrated among single-listers, and it *rises* with list length (Table 3), as expected if longer lists reveal stronger non-proximity preferences; single-listers gain not from an estimated taste but because deferred acceptance secures their stated school—usually their nearest, for 70% of them—against the displacement the distance lottery imposes. Third, and most directly, the headline welfare share is recovered when the average is taken over the credibly identified families alone—holding the full-sample parameter estimates fixed rather than re-estimating on the subsample: restricting to the applicants who rank more than two schools—the subsample for which γ_i is estimated (about 21, 16, and 12% of applicants in the three grades)—deferred acceptance recovers 99.3% and 90.3% of the attainable range in the two preschool grades (bootstrap 95% intervals [98.9, 99.6] and [84.1, 94.3]), essentially the head-

²⁰ Asked in the same survey why they did not rank additional schools, families overwhelmingly cite the scarcity of acceptable nearby options rather than missing information: 55% report that no public school is sufficiently close and a further 33% that they are confident of a seat at a listed school, while only 6% say it was difficult to find more schools—the sole information-friction response. Short lists thus reflect the local structure of acceptable options rather than an inability to locate alternatives.

line figure in Preschool 1 and *higher* in Preschool 2. The cardinal welfare object therefore is not an artifact of the structure imposed on single-listers. In congested Primary 1 the DC→DA gain itself nearly vanishes (0.012 km against an attainable range of 0.318), so the recovered share is imprecise—a bootstrap interval of roughly $[-77, 36]\%$ around the 3.9% point estimate—and is best read as “near zero” rather than a precise figure.

Table 3: DC→DA welfare gain by submitted-list length

List length	Preschool 1		Preschool 2	
	% appl.	gain (km)	% appl.	gain (km)
One school	57	+0.56	67	+0.15
Two schools	22	+0.70	17	+0.49
Three or more	21	+1.02	16	+1.03

Note: Mean per-family DC→DA welfare gain (km), posterior-mean utilities averaged over the 100 lottery draws, by the number of schools the family ranked. The gain rises with list length in both grades. Primary 1 is omitted: its total DC→DA gain is approximately zero (Table 8), so a decomposition is uninformative.

Table 10 in Appendix B presents the estimates from equation 1, and the potential scale reduction factors (Gelman et al., 1992) to assess mixing and convergence of the Gibbs sampling procedure (values close to one imply convergence). We also present the trace plots of the estimated σ_ϵ^2 in each iteration of the Gibbs sampling in Figure 10 of Appendix A. We discarded the initial 50,000 iterations of the Gibbs sampler as a burn period and used the following 100,000 to compute the mean parameters and standard deviations. The trace plots show that the values of σ_ϵ^2 remained stable. Estimates eliminating random coefficients from equation 1 are presented in Table 16 of Appendix B.1, and estimates of the main specification without siblings are presented in Table 17 of the same appendix section; the latter drops both the sibling term and the sibling-linked applicants, so it is estimated on a smaller sample. We observe that estimated parameters are very similar across the three alternative models, consistent with the similarity of the welfare estimates using the different specifications.

The model versus a distance-only benchmark. It is useful to read these estimates against the implicit model behind the status quo, in which families value only proximity. Adding the school effect ξ_j and a random coefficient on distance lets the model rationalize the families who do not rank their nearest school first; the welfare gain we report below (Table 8) is the distance by which the data reject this proximity-only model—had families valued only distance, there would be almost nothing for a preference-based rule to recover. The residual would not be exactly zero, since DC and DA differ also in priority protection and tie-breaking and those can move assignments even under $u_{ij} = -d_{ij}$; but it would be bounded by those design differences rather than by preferences, and the access decomposition suggests those differences are small on the margin it measures (Table 6). Because the size of that bound is an empirical matter that theory does not settle, we quantify it directly: we rebuild the market as the proximity-only model describes it, with utilities $u_{ij} = -d_{ij}$ and, in place of each submitted list, the distance-ordered list such a family would send, truncated at that family’s own observed length, and we

re-run every mechanism on that reconstructed market over the same 100 lottery draws.²¹ The entry-grade gain from the distance rule to deferred acceptance falls from 0.459 km under the estimated preferences to 0.0005 km, half a metre; it is -0.011 km in Preschool 2 and -0.003 km in Primary 1. The remainder is attributable to the design differences rather than to the elicitation: telescoping the null gain assigns 0.000 km to acting on the reported order, 0.0007 km to priority protection and -0.0003 km to tie-breaking. We conduct both arms on the simulation engine’s in-market sample, where the estimated preferences give a distance-rule-to-deferred-acceptance gain of 0.459 km against the 0.686 km implied by Table 8, and we therefore read the exercise as informative about the contrast between the two arms rather than about the level of either. Dropping the random coefficient, or the sibling term together with the sibling-linked applicants (Tables 16–17), changes neither the estimates nor the welfare conclusions, so the result rests on eliciting preferences, not on the particular parametric form. The two subsections that follow examine the model’s components in turn: what the school effect ξ_j captures and whether observed attributes can proxy it, how families substitute across schools, how the value of distance varies with family circumstance, and how that cost is best measured.

5.2 School desirability, and what a planner can observe of it

What explains school desirability? Consider what the school effect ξ_j captures. It is *not* the facilities we observe: giving ξ_j a hierarchical prior $\xi_j \sim N(X_j\beta, \sigma_\xi^2)$ on five facility indicators (information and science labs, internet, library, dining room) leaves all but one coefficient imprecise in the preschool grades and the residual dispersion in ξ_j large; in Primary 1 two indicators are significant, but the signal they carry is school size, and residualizing them on log enrolment removes most of it. As an exploratory step, given how few school attributes we observe, we measure richer characteristics from satellite and street-level imagery—building footprint, building count, built-up density, greenery, open recreation space, building condition, sidewalk quality, road quality, street-safety cues, and commercial activity—and re-estimate the same hierarchical model, so each β_k is a willingness to travel (km) for a one-standard-deviation increase in attribute k . The built form around the school carries a robust signal (Appendix Table 15). The two attributes that carry it are the total building footprint and the number of distinct buildings within 75 metres of the school, which describe the surrounding block, not the campus: families accept about 0.3 km less travel per standard deviation more buildings in that radius, significant in all three grades, and about 0.4 km more for a larger total footprint, a coefficient that is significant conditional on built-up density (the two correlate at 0.91) and imprecise in Primary 1 on its own. Schools in blocks of fewer, larger structures are more desirable. Going beyond hand-coded attributes, we also embed each school’s facade and interior photographs with a pre-trained vision model (Compiani et al., 2026) and enter the leading image dimensions directly as an additional desirability proxy; the leading learned dimension is itself worth roughly 0.2–0.3 km of travel per standard deviation (0.18 in Preschool 2) and clears a permuta-

²¹Ten arms in all, including positive controls and a permutation check; the arm we report replaces both the utilities and the lists. Replacing only the utilities leaves families submitting the lists they actually sent, which credits deferred acceptance with honouring rankings the proximity model says would never have been written; and because the distance rule is a serial dictatorship over distance (Lemma 1), such an arm asks little more than whether the distance rule minimizes distance. The substitution is material, not nominal: for 44% of entry-grade families the null list names a different set of schools than the one they submitted. We report kilometre levels and not shares of an attainable range, because under the null that range collapses to a few metres and a ratio with so small a denominator carries no information.

tion null in Preschool 1 and Primary 1 ($p = 0.003$ and 0.03 ; $p = 0.07$ in Preschool 2). The finer ground-quality attributes are individually imprecise, and most of the dispersion in ξ_j remains unexplained even with these richer measures—school desirability is real and largely idiosyncratic, which is why no parsimonious set of observable amenities can substitute for eliciting preferences directly. The attributes are generated regressors entered into the hierarchical prior and the reported intervals do not propagate the uncertainty in deriving them from images, so we read individual coefficients as indicative rather than definitive. The exercise is nonetheless a direct test of whether observable amenities substitute for elicited preferences, and they do not: with ten imagery attributes and learned image dimensions in the prior, most of the dispersion in ξ_j remains unexplained in sample, and leaving one school out at a time, essentially all of it does. School desirability is real, large, and largely unobservable to a planner, which is the measurement counterpart of this paper’s central claim: where recorded school attributes account for little of what families value, an assignment rule that does not elicit preferences has no observable proxy with which to replace them.

Differential substitution. Whether families also weigh these attributes *differently* from one another is harder to establish, and we do not claim it. Allowing per-family random coefficients on the two leading imagery indices returns dispersions of 0.2–0.3 km per standard deviation, but indices with no imagery content at all—permuted across schools, or pure noise—return the same band with credible intervals that likewise exclude zero, because the prior on the covariance keeps the dispersion away from zero by construction; only the first index in the preschool grades clears that placebo, and narrowly. A held-out exercise points the same way: conditional on distance and popularity, image similarity to a family’s first choice does not improve prediction of its second, and the in-sample coefficient is if anything negative. The value of the richer attributes is therefore in recovering the non-distance *level* of desirability that a proximity benchmark misses—which is what overturns “distance is all that matters”—not in predicting substitution. Consistent with the heterogeneity visible in Figure 1, the estimated desirability ξ_j is essentially uncorrelated with a school’s distance from the city centre yet strongly correlated with revealed demand—desirable schools are scattered across neighbourhoods, not concentrated downtown, which is why acting on preferences moves families across the map rather than simply closer to home.

Heterogeneity by family circumstance. Letting the distance coefficient vary with observed neighbourhood socio-economic status—the average adult schooling of the home census block (2022 Census)—shows that families in higher-schooling neighbourhoods are *more* distance-averse: the random coefficient γ_i falls with neighbourhood education on the full applicant sample, and the same pattern holds using mother’s education from the companion survey. Better-off families are also better located relative to desirable schools, so they have less to gain from a move toward their stated preferences. That location channel is one route to a progressive incidence, but it is not the route operating where we find one: in Preschool 2, where the gradient is robust, block schooling is essentially uncorrelated with distance to the nearest school and most of the contrast survives residualising on geography. In Preschool 1, where the location channel is present, the contrast does not survive it.

5.3 Measuring travel cost, and what the linear specification misses

The near-universal specification in this literature, utility linear in straight-line distance, is misspecified: a concave cost in *walking* time, $\log(1 + d)$, predicts the submitted lists 485 held-out log-points better than linear straight-line kilometres on the 1,973 ranked positions of Preschool 1 applicants, and it is the best-fitting cost in the two later grades as well (Table 4). Two consequences follow, of which the second is not particular to this setting. The advantage that drive time appears to hold over crow-flies distance is curvature, not the road network, since once the cost is allowed to bend walking time fits better than driving time. And a third to a half of the dispersion in the distance coefficient σ_γ is absorbed by curvature alone, so that the σ_γ of a linear-in-distance specification is better read as an upper bound on heterogeneity in tastes for travel than as a measure of it.

Travel time as the cost measure. The estimator and the status-quo rule both use straight-line distance, and it mismeasures the cost families face—not mainly because Manta’s hills, water-front and trunk highway bend routes, which adds a fairly uniform detour of about $1.4\times$ in road distance, but because the pace of a trip is not constant in its length. Replacing straight-line distance with Google-calibrated drive time leaves the correlation high (about 0.95, and 0.92–0.96 across grades and populations), yet among Preschool 1 applicants’ in-market home-school pairs 28% take more than $1.5\times$, and 7% more than twice, the time a constant minutes-per-kilometre rate implies. Almost all of those pairs are short: a fixed access cost of roughly four minutes means trips under a kilometre run at about 13 km/h and trips beyond ten kilometres at about 33, so a cost linear in crow-flies distance is wrong precisely over the range where most applicants live from most schools. The exercise changes only how the assignments are *valued*: we hold the realized DC and DA assignments fixed at the same 100 lottery draws and re-estimate demand with drive time in place of straight-line distance, so the status-quo rule still allocates on straight-line distance, as it did in practice. To keep the two estimates comparable we put minutes on a kilometre-equivalent scale, dividing drive time by the through-origin slope of time on straight-line distance (its distance-weighted average minutes per kilometre); welfare under either measure is then denominated in those same units, so the headline figures elsewhere in the paper remain straight-line kilometres. Re-estimating on that common scale leaves λ and σ_ϵ within a few percent of their straight-line values but cuts the distance-taste heterogeneity σ_γ by roughly a third to a half in every grade. The comparison is made on matched 12,000-draw chains, where the reduction is 30–44%; at 40,000 draws it is 39–49%, and σ_γ is identified from the 226, 147 and 46 applicants who ranked more than two schools. The welfare gain itself is essentially unchanged in the entry grades (within about one percent), since it derives from eliciting preferences, not from how distance is measured; in congested Primary 1 the gain is small and imprecisely estimated, and the ratio between the two cost measures is correspondingly noisy.

Curvature rather than the road network. The reason drive time helps is not that it carries information about routes that straight-line distance lacks. Drive time is close to a concave function of straight-line distance, and a concave transform of straight-line distance itself— $\sqrt{d}$ or $\log(1 + d)$ —delivers the same reduction in σ_γ . Held-out prediction of the submitted rankings separates the two explanations (Table 4). Entered linearly, drive time does beat straight-line

Table 4: Which cost measure and which school covariates best predict the submitted rankings

	Preschool 1	Preschool 2	Primary 1	Param.
<i>Panel A. Cost measure, with no school covariates</i>				
Straight-line distance (<i>main specification</i>)	—	—	—	2
Drive time	+303.2	+185.9	+51.7	2
Walk time	+97.2	+54.1	+29.6	2
$\log(1 + d)$, straight-line distance	+474.7	+312.2	+120.0	2
$\log(1 + d)$, drive time	+441.9	+294.7	+96.5	2
$\log(1 + d)$, walk time	+485.2	+362.1	+126.0	2
<i>Panel B. School covariates, added to $\log(1 + d)$, walk time</i>				
Enrolment and facility indicators	+637.7	+441.0	+204.8	8
Imagery attributes	+633.6	+445.7	+220.4	12
Image embeddings	+597.5	+365.1	+135.7	10
Imagery attributes and embeddings	+653.2	+461.5	+227.3	20
Facilities and imagery attributes	+667.6	+453.7	+236.1	18
All three blocks	+678.7	+484.4	+251.3	26
Unrestricted school effects	+772.1	+521.0	+193.7	56
Families	1,098	885	389	
Ranked positions	1,973	1,437	599	
Programmes	55	57	54	

Note: Held-out log-likelihood, relative to the paper’s linear straight-line cost; higher is better prediction of the rankings families actually submitted. Exploded rank-ordered logit on each submitted list, choice set all in-market programs of the grade, five-fold cross-validation splitting on *students* so that every school appears in both training and test folds and school-level covariates and unrestricted school effects are alike estimable and transferable. Seed 20210312. The specification is $V_{ij} = -\beta f(c_{ij}) + \lambda \text{sib}_{ij} + Z_i'\theta$; the sibling term is present throughout. The *Param.* column counts estimated parameters on the Preschool 1 basis and includes the cost and sibling terms, so the twenty-four school covariates of the last Panel B row enter as 26 and Preschool 1’s fifty-four unrestricted school effects as 56; the other grades have 57 and 54 programs. Panel A varies only the cost. The three *linear* rows are invariant to the units of the cost variable, since β absorbs any rescaling, so the comparison among them needs no convention. The $\log(1 + d)$ rows are not scale-invariant, so each measure is first placed on a kilometre-equivalent scale—multiplied by $\bar{d}/\bar{x}$, which preserves zero—before the transform; otherwise the amount of curvature would differ across measures purely because minutes are larger numbers than kilometres. Panel B adds each school-level block to the cost that wins Panel A, and the blocks are alternatives, not a nested sequence: the last three combine them. Two readings require care. The cost is selected on the same folds against which the gains are then reported; with six candidates and differences of this size that will not reverse any ordering here, but the pre-specified contrast is linear drive time against linear straight-line distance. And the gap to *unrestricted school effects* measures out-of-sample predictive efficiency at this sample size, not the share of desirability that observables explain: with one or two ranked schools per family the unrestricted school effects are themselves noisily estimated, which is why in Primary 1—whose 54 programs give fifty-three unrestricted school effects—a model with twenty-four school covariates out-predicts them. That row is therefore a comparison, not a ceiling. The imagery attributes and image embeddings are rebuilt here from the attribute file and the saved embeddings rather than from the design matrix behind Table 15. Two schools in each of the later grades have no photographs and their principal components are mean-imputed, which if anything understates what the image blocks contribute there. Source: authors’ estimation, `code/4_analysis/heldout_fit_specs.py`.

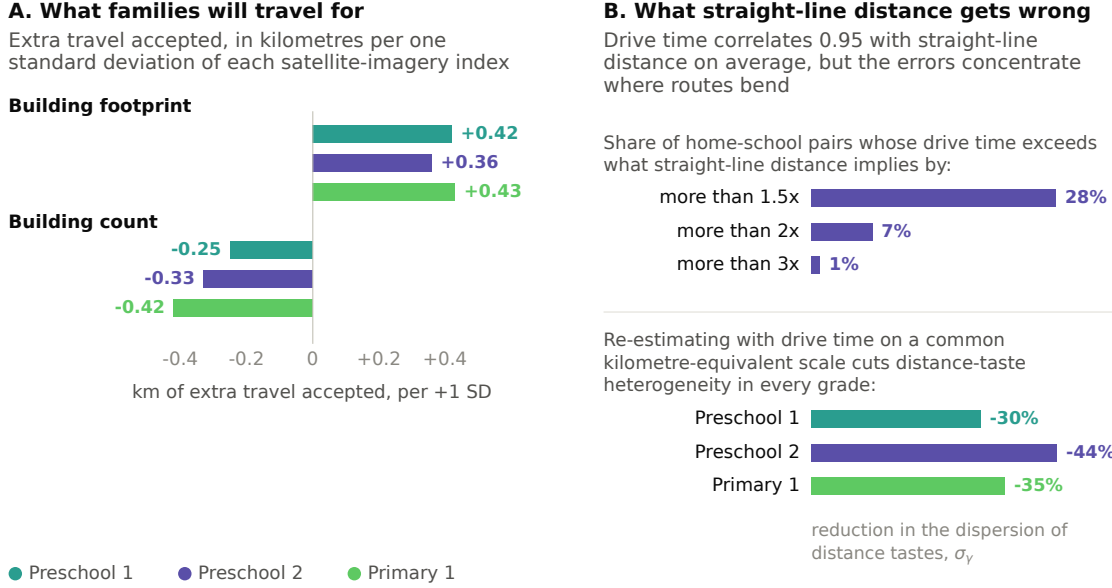
distance, and by a wide margin. But once the cost is allowed to bend, $\log(1 + d)$ in straight-line distance beats drive time in every grade, and the best-fitting cost of all is $\log(1 + d)$ in *walk* time—which is close to network distance, since walking speed is nearly constant. The street network does therefore carry something beyond crow-flies distance, but it is recovered by walking time under a concave cost, not by driving time; what the linear specification was missing is mostly curvature, which the random coefficient had been absorbing.

We nonetheless retain straight-line kilometres in the main specification. It is the metric the status-quo rule itself acted on, and the linear normalization is what denominates our welfare estimates in kilometres: under a concave cost one unit of utility is worth $(1 + d)$ kilometres at distance d , so no single conversion exists. Table 4 is a statement about fit, not a proposal to re-base the welfare numbers—and Figure 12 reports what the headline does when the best-fitting specification is used instead: Preschool 1 does not move and Preschool 2 rises, both well inside the spread the other specifications already span.

Figure 3 collects what the richer data buy. Panel A reports the willingness to travel implied by each imagery index; Panel B, the share of home–school pairs whose drive time departs from what straight-line distance implies, and the reduction in σ_γ that follows from re-estimating on the drive-time scale.

Both measurement exercises point in the same direction. Better data improve what a planner can measure—the shape of travel cost in the one case, the level of a school’s desirability in the other—without making either fully observable. Imagery and embeddings recover the non-distance *level* of desirability that a proximity benchmark misses; they do not identify how families substitute across schools. Even with the facility indicators, ten imagery attributes and learned image dimensions all in the prior, most of the dispersion in ξ_j is unexplained in sample, and school by school essentially all of it is. We read that residual as the measurement case for the design this paper evaluates: the component of school desirability that no set of observables we assemble recovers can enter the assignment only if families are asked to report it.

Figure 3: What better data on schools and on travel cost buy



Note: Panel A: willingness to travel, in kilometres per one standard deviation of each satellite-imagery index, by grade (Table 15). Families accept measurably less travel for a higher count of buildings within 75 metres of the school and more for a larger total footprint in that radius—the built form of the surrounding block, not the campus—so schools are not interchangeable conditional on distance—which is what a proximity rule assumes. Panel B, upper: the share of Preschool 1 applicants’ in-market home-school pairs whose Google-calibrated drive time exceeds a constant minutes-per-kilometre rate, by factor; almost all such pairs are short trips, reflecting a fixed access time, not route detours. Panel B, lower: the reduction in the dispersion of distance tastes σ_γ when demand is re-estimated with drive time on a common kilometre-equivalent scale. The reduction is a within-experiment comparison at a shorter chain length, both arms estimated identically; the levels are therefore not comparable with the σ_γ of Table 10, which is why only the reduction is shown. Source: authors’ estimation.

As in [Abdulkadiroğlu et al. \(2017\)](#), we estimate utilities conditional on the estimated parameters and, importantly, also conditional on the reported preference rankings:

$$\mathbb{E} [u_{i,j} | r_i, \xi, \lambda, \sigma_\epsilon, \sigma_{\tilde{\xi}}, \Sigma_\gamma, d_i]$$

Here, r_i is i ’s reported preference ranking, and to compute i ’s expected utility if assigned to school j we average directly over the iterations of the Gibbs sampler, which lets us condition easily on reported preference rankings. Estimated average utilities are measured in kilometres (km), which is a feature of using a scale normalization of -1 on the linear distance parameter.

6 Results

We first describe the differences between systems in assignment to preferences and linear distance to home, then present our welfare comparison using the utility model introduced above.

Notably, 55.97% of applicants (n=2,225) ranked first the very program their sibling-adjusted distance ordering places first—the input the distance rule acts on—underscoring how families

value proximity. This identity-based share is stricter than the distance-based complement of Figure 2’s 28%, which tolerates ties and sub-100-metre differences, so the two measures deliberately differ. Table 7 compares the DA and DC algorithms, averaging outcomes over 100 lottery draws at a fixed seed. Although the DC alternative assigned fewer applicants to their preferred school overall, the percentages were similar for those prioritizing distance, as shown in rows 1 and 2 of Panel B. This indicates that a coordinated system incorporating preferences does not disadvantage applicants focused on proximity. However, the DA algorithm assigned students to schools an average of 0.32 km farther away than the DC algorithm.

A defining property of deferred acceptance is that it produces a *stable* assignment: no applicant prefers a school to her own placement while holding higher priority there than an admitted applicant. The distance mechanism does not. Evaluating each assignment against families’ reported preferences and the priority structure of the choice system, the DA assignment leaves *zero* applicants with justified envy (by construction), whereas under DC 44.9% of applicants (1,784) have justified envy — they reported a school they prefer to their distance-based placement and hold higher priority there than someone admitted to it — and 45.2% belong to some blocking pair (Table 5). The diagnostic evaluates the DC allocation against the *reformed* priority structure, not against DC’s own common processing order, so it combines the change in what the algorithm ranks with the change in priorities and tie-breaking; it measures how the old allocation fares by the new system’s standard, not instability caused by ignoring preferences alone. This is the formal counterpart of the school-change-petition “aftermarket” described in Appendix C—a petition phase that our simulated distance counterfactual does not reproduce, so the reported gain is an upper bound to that extent—and it is close to the 44.0% of applicants whose reported first choice is not the first program of their own sibling-adjusted distance ordering—the deterministic object the distance rule feeds on, defined before any lottery is drawn—the complement of the 55.97% who ranked that program first.²² Because the within-grade rates are essentially flat across Preschool 1, Preschool 2 and Primary 1 (Table 5), instability under the distance rule is pervasive at every level of congestion.

6.1 Where the Gain Comes From: Eliciting Preferences vs. the Algorithm

The move from DC to DA changes three things at once: it acts on *reported* preferences rather than distance, it gives reported programs higher priority than the distance-imputed fallback, and it breaks ties with a separate lottery at each school (multiple tie-breaking). To see which of these drives the large increase in assignment to preferences, we decompose the gain by walking from DC to DA one design lever at a time, holding the lottery draw fixed (Table 6; means over 100 draws). *RSD-P* is a serial dictatorship: it runs over reported-then-distance lists with the flat DC priority and a single tie-break, so it differs from DC only in using reported rather than distance order. *RSD-P-PRIO* then adds the reported-over-imputed priority. That makes priority

²²This 44.0% is a related but distinct measure from the 28% in Figure 2: the latter counts families whose stated first choice lies more than 100 metres beyond their nearest *available* school (median +0.59 km of additional travel), whereas the 44.0% counts any first choice that is not the *identical* first program of the distance ordering—so it additionally sweeps in ties, sub-100-metre differences, and different programs at comparable distance. Both are computed *ex ante*, from each family’s distance ordering over the programs available to it together with the sibling adjustment, not from any one realized lottery draw; they differ because the second conditions on the single program the rule would hand that family, while the first only asks whether the stated first choice is nearer or farther than the closest option. We use the more conservative 28% as the headline descriptive fact.

Table 5: Stability: Justified Envy and Blocking Pairs

	DA	DC
Applicants with justified envy	0 (0.0%)	1,784 (44.9%)
Applicants in a blocking pair	0 (0.0%)	1,797 (45.2%)
<i>Justified envy under DC, by grade (% of that grade's applicants)</i>		
Preschool 1		739 (43%)
Preschool 2		783 (46%)
Primary 1		262 (46%)

Note: An applicant has justified envy if she prefers a school to her assignment (by her reported preferences) and has higher priority there — under the deferred-acceptance priority structure — than an admitted applicant; a blocking pair additionally includes preferring a school with a free seat. DA is stable by construction. DC figures are for a single lottery draw. The two upper rows report shares of all applicants; the by-grade rows report shares of that grade's applicants.

school-specific, so it is no longer representable as a single processing order and is implemented as deferred acceptance under a common lottery; DA in turn adds multiple tie-breaking. The decomposition therefore walks through design levers, not through one mechanism class, and the incentive and stability properties attach to the final step, not to the intermediate ones.

The decomposition is stark: of the 28-point increase in the share assigned to a listed preference (50% to 78%), **23 points — more than four-fifths — arrive at the first step, when the algorithm begins acting on reported order**, a further 5 points when reported programs gain priority over imputed ones, and essentially none (-0.04) at the final step, multiple tie-breaking. The walk is sequential, so these are path attributions—each lever is measured at the allocation the previous levers produced rather than as an order-independent share—but the first step dwarfs the rest from the natural baseline, the status quo. The pattern is the same for first-choice assignment, where multiple tie-breaking in fact slightly *reduces* the share (by 0.7 points), trading a few first-choice placements for the stability and strategy-proofness that DA delivers. The headline “access to preferred schools” gain is thus overwhelmingly about *asking* families what they want; the value added by the matching algorithm itself lies not in this margin but in the incentive and stability properties documented above (Proposition 1, Table 5) and in the welfare comparison below.

The decomposition in Table 6 is an access decomposition. Telescoping the welfare gain through the same three levers, on the simulation engine's in-market pool and on a fixed lottery draw, gives attributions that differ sharply across grades. In Preschool 1, acting on the reported order accounts for 0.4581 of the 0.4593 km gain, or 99.7%; in Preschool 2 for 0.1956 of 0.3246 km, or 60.2%; and in Primary 1 for 0.1046 of 0.3141 km, or 33.3%, with the protection of reported programs over the distance-imputed fallback accounting for 96.9% and tie-breaking subtracting the remainder. The levels are those of the in-market pool, on which the entry-grade gain is the 0.459 km reported in Section 5.1 rather than the 0.686 km implied by Table 8, and the legs are measured sequentially, so they are path attributions and need not lie between zero and one. We therefore scope the claim that the gains come from eliciting preferences to the entry grade: in the later-entry grades most of what deferred acceptance delivers is the priority that reported programs hold over the imputed fallback, which acting on the reported order makes possible

but which is not the reordering itself.

Table 6: Decomposition of the Assignment-to-Preference Gain (DC $\rightarrow$ DA)

	Any preference	First preference
DC (distance)	50.1%	42.3%
+ elicitation (reported order)	73.1% (+22.99)	66.6% (+24.30)
+ priority protection	78.3% (+5.18)	70.4% (+3.74)
+ multiple tie-breaking = DA	78.2% (−0.04)	69.7% (−0.65)
Total DC $\rightarrow$ DA	+28.13	+27.39

Note: Share of applicants assigned to a reported preference, mean over 100 lottery draws, decomposing the DC $\rightarrow$ DA gain by changing one design feature at a time (margin in parentheses). The intermediate mechanisms are serial dictatorships over reported-then-distance lists (see text).

Table 7: Mechanism Comparison - Results

	DA	DC
Panel A: Applicants assigned in:		
<i>Any preference</i>	3,109 (78.22%)	1,990 (50.07%)
<i>First preference</i>	2,770 (69.68%)	1,682 (42.32%)
<i>Average assignment distance</i>	1.33 km	1.01 km
Panel B: Applicants with the same 1st preference (2,225) assigned in:		
<i>Any preference</i>	1,789 (80.40%)	1,674 (75.24%)
<i>First preference</i>	1,657 (74.47%)	1,621 (72.87%)

Note: Assignment outcomes under deferred acceptance (DA) and the distance rule (DC), averaged over 100 lottery draws at a fixed seed (95% across-draw intervals are within ± 0.7 percentage points): all 3,975 applicants in the assignment files (Panel A; the 9 applicants with special educational needs are not in them) and the 2,225 applicants whose first preference coincides with their distance-priority school (Panel B). Population shares in parentheses.

Figure 4 shows the rank of the assigned school within each applicant’s submitted list for both systems. As we can see, the DA algorithm assigns more students to their first preference than the DC algorithm (70% to 42%) and far fewer students to an alternative outside of their reported preference list (22% vs 50%). Tables 11 to 13 in Appendix B and Figure 9 in Appendix A display these results for the different grades. Greater congestion leads to smaller differences between the two mechanisms in applicants assigned to their preferred options: more congestion both lowers the share assigned to a reported preference under DA and, under the distance rule, raises the chance that an applicant placed in a closer school displaces another who ranked that school first.

As a final robustness check, we bound the effect of any location misreporting under the

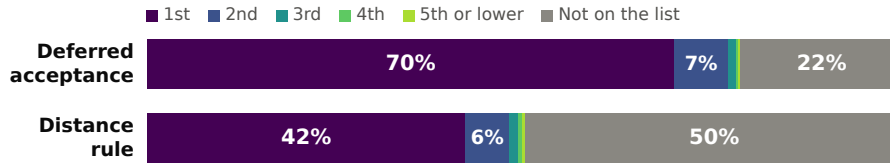


Figure 4: Rank of the assigned school within the applicant's submitted list

Note: Share of applicants assigned to their first, second, third, fourth, or fifth-or-later listed school, or to a school they did not list ("Not listed"), under deferred acceptance (DA) and the distance rule. Pooled across grades, single representative lottery draw at the fixed seed. Deferred acceptance raises first-choice placement from 42% to 70% and reduces placement outside the submitted list from 50% to 22%. Source: authors' calculations.

distance rule. Even if up to 90% of applicants were to report an address next to their most preferred school, the share assigned to a listed preference would rise only to about 61%—still well short of the 78% that deferred acceptance achieves—because a falsified address can signal a preference for at most one school (Table 21, Appendix B). Two limits on what this delivers. It is a stress test under one manipulation rule—misreporters are moved next to their stated first choice—rather than a bound on optimally chosen manipulation, which could target a less congested listed school instead. And it bounds *placement in a listed school*, not mean utility: welfare also depends on which listed school is obtained and on travel from the true home. We therefore read it as showing that the access gap does not close under heavy misreporting, not as establishing that the cardinal welfare comparison is insensitive to it.

6.2 The welfare ranking is not an artifact of the metric

Because we measure welfare in the very preferences deferred acceptance is designed to satisfy, one might think DA must win by construction. But the object of interest is not the *sign* of the gain but its *magnitude*. The sign is not guaranteed by theory: deferred acceptance neither nests nor Pareto-dominates the distance rule, and with binding capacities, reported-over-imputed priority and school-specific tie-breaking, individual applicants can lose and the mean could in principle fall, so that it does not is itself an empirical finding. The magnitude is what is falsifiable: had families valued only proximity, so that the distance rule's implicit model of preferences were correct, preference-based assignment would have recovered essentially none of the attainable welfare. Imposing that proximity-only model—distance utilities, and the distance-ordered lists such families would then submit, truncated at each family's observed length—collapses the entry-grade gain from the distance rule to deferred acceptance to 0.0005 km, against the 0.459 km the estimated preferences give on the same in-market basis (Section 5.1). The estimated gain therefore measures how far revealed preferences depart from the proximity presumption, disciplined by the descriptive fact that more than a quarter of families rank a school other than their nearest first (Figure 2). Three features bound how much of the gain can be mechanical. First, the gain is not tied to the deferred-acceptance objective: the decomposition above attributes essentially all of the *access* gain to eliciting preferences—more than four-fifths to acting on the reported order, the rest to the priority protection that order makes possible—and essentially none to the deferred-acceptance machinery (Table 6)—while

deferred acceptance closes nearly all of the attainable range (Table 8) and the efficiency refinements beyond it add only 0.003 km in the entry grade. The first is an access decomposition, not a welfare one, and it is pooled across the three grades where the welfare figures in this subsection are Preschool 1; what links them is that the welfare left unclaimed *after* acting on preferences is negligible, so the gain cannot be a reward to the deferred-acceptance machinery in particular. Telescoping the welfare gain through the same levers, as we do in Section 6.1, attributes almost all of the entry-grade gain to acting on the reported order and a substantially smaller share of it in the later-entry grades. Second, the recovered share is far from uniform: it is much smaller in the congested grades—99.6% in Preschool 1 against 82.3% in Preschool 2, on non-overlapping intervals, and near zero in Primary 1 (Appendix B.1). That is what a congestion channel predicts—though the grades also differ in applicant composition, so the gradient is consistent with the economic primitive, not proof of it—and an advantage built into the metric alone would have no reason to track seats. The design levers, however, do track seats, so the gradient narrows a mechanical reading rather than excluding it. Our simulation engine draws the tie-breaking lottery independently for each applicant–school pair, which is finer than the rule the system deployed and therefore bounds the tie-breaking leg in magnitude; setting that leg to zero leaves a preference-free bound on the proximity-only gain that rises with congestion instead of falling, from 0.0007 km in Preschool 1 to 0.014 km in Preschool 2 and 0.095 km in Primary 1, against the 0.459 km the estimated preferences deliver in the entry grade on that same in-market basis. Third, our welfare metric is not a free-floating utility index but willingness to travel, denominated in kilometres—the status quo’s own currency. The additional travel deferred acceptance induces, 0.33 km per applicant in the entry grade, is already netted out within it: the 0.69 km gain is what families’ revealed valuations of their preferred schools are worth *after* paying that distance, in kilometres they themselves accept. What remains is the generic caveat that any cardinal welfare statement rests on the estimated utility scale (Abdulkadiroğlu et al., 2015), which we take up with the other limits of the exercise in Section 6.4.

6.3 Welfare in kilometres of willingness to travel

Table 8 reports the estimated differences in mean utilities (both in km) and in standard deviations relative to a constrained-efficient (TTC) assignment benchmark.²³ In Panel A, differences between the DA and TTC algorithms are small in welfare terms (less than 80 metres) at the preschool levels, and significantly smaller than the welfare loss under the DC (distance) alternative (0.689 km and 0.430 km on average, respectively). Equivalently, of the welfare distance between the distance rule and the constrained-efficient benchmark, deferred acceptance closes 99.6% in Preschool 1 (one minus the DA gap divided by the DC gap in Panel A). The statistic is a share of *that* range, not of the largest mean utility the market could produce: the benchmark is Pareto-efficient subject to priorities and is reached from deferred acceptance by ordinal trades, so it is a priority-respecting ceiling, not a cardinal one. To see how much that distinction matters we computed the cardinal ceiling directly: the capacity-feasible assignment of these same estimated utilities that maximizes mean welfare, ignoring priorities and stability altogether.²⁴ It reaches 0.681 km in Preschool 1 against the constrained benchmark’s 0.586, and

²³SIC and TTC actually have the exact same assignment in all 100 simulations, as explained in Appendix E.

²⁴An exact maximum-weight assignment on the estimated utility matrix, with each program expanded to its seats in the assignment code’s own capacity file, solved by rectangular linear assignment. It is an upper bound no priority-respecting mechanism can attain, so it brackets rather than replaces the benchmark in Table 8.

–1.072 against –1.433 in Preschool 2. Measured against that unconstrained ceiling, deferred acceptance closes 87.6% of the range in Preschool 1 and 44.7% in Preschool 2 (applicant-level bootstrap 95% intervals [85.3, 89.8] and [36.4, 52.7], constructed as for the headline share). That ceiling is computed at the posterior-mean utilities, so it is the most a planner with our information could attain; a planner who also observed each family’s realized taste shocks could attain more on average, and against that higher, draw-by-draw ceiling deferred acceptance closes correspondingly less.²⁵ The welfare-maximizing assignment that *does* respect priorities lies between the two ceilings, so wherever it falls, deferred acceptance closes between 87.6 and 99.6% of the relevant range in the entry grade and between 44.7 and 82.3% in Preschool 2, where both endpoints depend on the valuation of out-of-market placements noted under Table 8. The ordering we report is unaffected—the distance rule forgoes far more than deferred acceptance under either ceiling—but the headline share is a statement about the priority-respecting range, and we read it that way throughout. An applicant-level bootstrap places that share in [99.5, 99.8].²⁶ The later-entry grades recover smaller shares (82.3% in Preschool 2 and near zero in congested Primary 1); both are less precisely identified, reflect a selected mid-stream applicant population, and are discussed in Appendix B.1. Regenerating the estimation–assignment pipeline end to end—re-estimating preferences at the full Gibbs chain and re-running the assignment under 1,000 fresh lotteries—reproduces the Preschool 1 share to within a tenth of a point (99.5%, against the 99.6% headline) and leaves the recovered shares unchanged to within half a percentage point under alternative variance-component priors, so the welfare object depends on neither the priors nor a particular set of lottery draws. The difference is larger among applicants assigned to different schools under the two algorithms, as shown in Panel B.

In Primary 1, the difference between the DA and DC algorithms is smaller because increased congestion places more applicants in a non-preferred alternative, as shown in Figure 9 of Appendix A. Moreover, Table 14 of Appendix B shows that the share of applicants assigned to the same non-preferred school under both algorithms rises sharply in Primary 1 (56.65% of applicants assigned to the same school, compared to 2.61% and 21.17% in Preschool 1 and 2 respectively). Furthermore, conditional on having a different assignment in the DC and DA algorithms, the share of applicants who move from a non-preferred to a preferred assignment under the DA algorithm is 41.12% in Primary 1, compared to 75.27% in Preschool 1 and 53.12% in Preschool 2 (i.e., the share of applicants with improved outcomes is smaller in later years).

²⁵Re-solving the maximum-weight assignment at each of 400 draws from the posterior of the demand parameters (160 in Preschool 2), taken from four independent chains of the sampler, gives a ceiling that is higher on average than one held fixed at the posterior mean—by 0.04 km in Preschool 1 and 0.38 km in Preschool 2, as Jensen’s inequality requires—and against it deferred acceptance closes 84.1% of the range in Preschool 1 (posterior 95% interval [82.9, 85.2]) and 29.3% in Preschool 2 ([26.4, 31.6]). The two are different estimands, not a biased and a corrected version of one: the first is the ceiling a planner with our information could reach, the second the average ceiling a planner who also saw each family’s taste shocks could reach. The headline share is unaffected by the choice, because its benchmark is the assignment engine’s output on the submitted lists, priorities and lottery draws, not an assignment chosen on the estimated utilities.

²⁶Percentile intervals from 2,000 resamples of the applicant pool, holding the estimated preference parameters and the simulated lottery draws fixed; they therefore describe sampling variation in the composition of the applicant pool, not posterior uncertainty in the demand parameters. The same procedure gives [77.2, 86.0] in Preschool 2 and [–77.0, 35.7] in Primary 1, where the ratio is a quotient of two congestion-compressed quantities and is best read as “near zero” rather than as a point estimate. Posterior 95% credible intervals for the same share, from four independent chains of the Gibbs sampler with the applicant pool held fixed, are [99.5, 99.7] in Preschool 1 and [79.4, 84.6] in Preschool 2, in line with the bootstrap; in Primary 1 the posterior, like the bootstrap, is too wide to locate the ratio.

Table 8: Differences in Welfare: Constrained-Efficient (TTC) Benchmark vs. DC and DA Algorithms

Measure	Preschool 1		Preschool 2		Primary 1	
	DC	DA	DC	DA	DC	DA
Panel A: All simulated applicants						
Δ Mean utility (km)	-0.689	-0.003	-0.430	-0.076	-0.318	-0.306
$\frac{\Delta \text{Mean utility}}{\sigma_{U_i, \text{TTC}}}$	-0.699	-0.003	-0.158	-0.028	-0.094	-0.090
% of DC–TTC range recovered	—	99.6	—	82.3	—	3.9
Panel B: Applicants with different assignments across algorithms						
Δ Mean utility (km)	-1.486	-0.005	-0.666	-0.118	-0.417	-0.402
$\frac{\Delta \text{Mean utility}}{\sigma_{U_i, \text{TTC}}}$	-1.466	-0.005	-0.263	-0.046	-0.127	-0.121

Δ Mean utility (km) is measured computing $u_{i,j(\mu)} - u_{i,j(\text{TTC})}$, where $j(\mu)$ represents the school to which individual i is assigned under mechanism μ . We then compute average utilities for each algorithm and simulation and finally compute the average for each algorithm across simulations. $\frac{\Delta \text{Mean utility}}{\sigma_{U_i, \text{TTC}}}$ simply uses the standard deviation of utility under the TTC mechanism to scale this difference in each simulation. This is done to facilitate extrapolations to other contexts. The final row of Panel A reports the share of the DC-to-TTC welfare range recovered by DA, computed as $1 - \Delta_{\text{DA}} / \Delta_{\text{DC}}$. This same table, without the final share row, is presented in Appendix B.1 for the specifications without random coefficients and without siblings. Where a mechanism places an applicant in a program outside the modelled in-market set, the placement is valued at the estimated utility of the in-market program occupying the same position in the applicant’s distance-appended list (a positional stand-in); in 84–99% of such placements the program is another shift (*jornada*) of a school that is in the modelled set. The Preschool 1 rows of this table are insensitive to that convention (the recovered share moves by at most 0.3 points under alternative valuations of such placements), whereas the later-grade levels are sensitive to it, which is one reason the headline rests on the entry grade.

Finally, the DC algorithm assigns schools closer to home—a feature of not prioritizing reported preferences—and since utility is on average greater at shorter home-to-school distances, this narrows the average utility difference between mechanisms.

Figure 5 shows the pooled distribution of estimated welfare under the two mechanisms. Deferred acceptance concentrates mass at higher utility—the median applicant sits at 0.01 against -0.95 under the distance rule, and 50% rather than 35% of applicants land above zero—while the distance rule holds visibly more mass around -2 km. The two far left tails are nearly identical, so the gain shows up in the bulk of the distribution, not among the worst-served applicants. Figure 11 of Appendix A breaks the same object out by grade and adds the TTC and SIC benchmarks, with two peaks in each figure: one among applicants assigned to a preferred option and another for those assigned to a non-preferred option close to home. The DA and TTC (and SIC) algorithms have very similar distributions. However, TTC does improve the assignment relative to DA in Primary 1, because with higher congestion the constraints that strict lottery tie-breaking imposes bind more tightly. SIC relaxes exactly those, remaining stable under the underlying weak priorities; TTC then executes any remaining trades without preserving stability. The two coincide here, and either achieves a significant improvement (0.306 km overall over the DA assignment and 0.402 km if restricted to applicants with different assignments), as shown in Table 8.

What the appended margin does to the refinement. The refinement band deserves a qualification we state plainly, because in the congested grades it is largely a property of how the

counterfactual lists are constructed. The mechanisms run on each applicant’s reported list with the remaining programs appended in order of distance, as the pilot’s own fallback did. Every appended program ranks below every reported one and carries the same, lowest priority, so no discordance of the sibling-priority kind arises on that margin. But a common priority is a tie, and stable improvement cycles exist to exploit ties (Erdil & Ergin, 2008). Classifying every move SIC makes in the simulations, 93% are trades between two appended programs: none in Preschool 1, where 93% of applicants already hold their stated first choice and there is nothing to trade, but 86 of the 88 applicants SIC moves per draw in Primary 1. The simulations also draw the lottery independently for each appended program, whereas the pilot used a single number across an applicant’s appended list (Section 4); under the pilot’s own rule the appended-margin trades disappear. Primary 1’s 0.306 km refinement is therefore not a lower bound on the scope for beneficial exchange but largely an artefact of two modelling choices on the appended margin, and we read it as such; in the entry grade neither choice matters, which is one more reason the headline rests there.

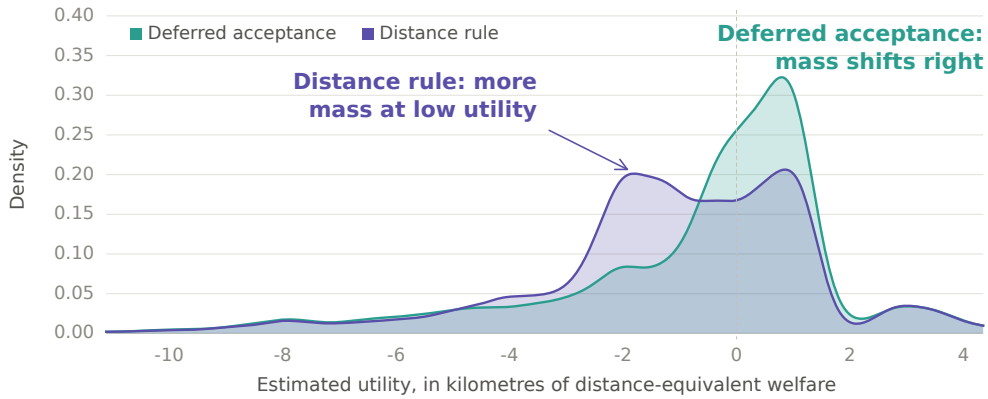


Figure 5: Distribution of estimated welfare under deferred acceptance and the distance rule

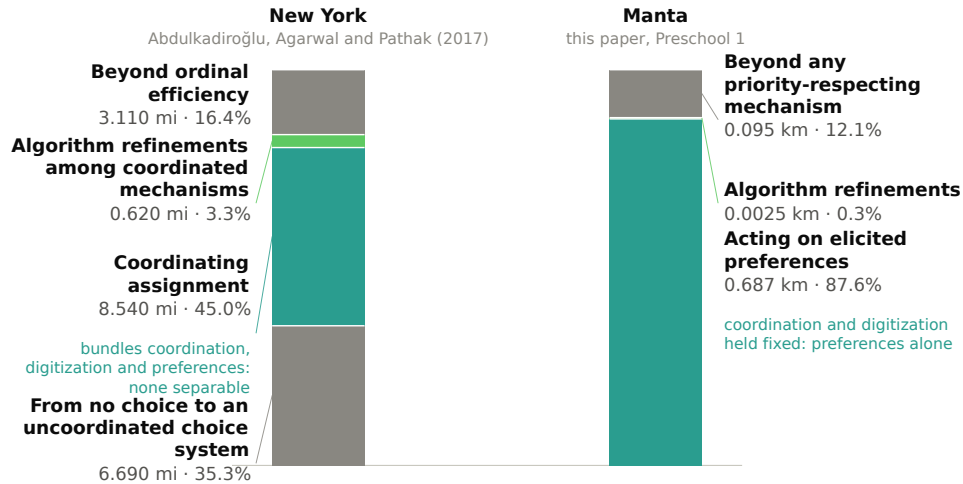
Note: Density of estimated applicant utility, in kilometres of distance-equivalent welfare, under deferred acceptance and the distance rule, on the 2,200-applicant welfare-comparison sample (Table 2) rather than the full pool of 3,984, since only that sample has modelled utilities. The utility level reflects the normalizations $\bar{d} \equiv 0$ and -1 per kilometre of distance and is not itself meaningful; the rightward shift of the deferred-acceptance distribution is the welfare gain. Source: authors’ estimation.

Figure 6 places the two settings side by side, redrawing the New York decomposition from the values reported in their Figure 5 on a common normalized scale. The two large bands are not the same object, and that is the point. New York’s move *out of* an uncoordinated system bundles coordination, digitization and acting on preferences, so none of its 45.0% can be attributed to any one of the three; Manta held the first two fixed and changed only the objective, so its 87.6% is acting on preferences alone. The band that share measures is 0.687 km: deferred acceptance minus the distance rule, which the rounded entries of Table 8 give as 0.686. That table instead reports each mechanism’s gap to the constrained-efficient benchmark, 0.689 km for the distance rule. The two settings are complementary, not a replication: ours identifies the component theirs cannot separate. What does carry across is the residual. Refinement among coordinated mechanisms is small in both—3.3% of the New York range and 0.3% of ours—so in neither setting does the welfare lie in the choice among coordinated algorithms: just as Ab-

Figure 6: Coordination and preferences: two settings that separate different things

The two decompositions, side by side

Each range normalized to 100%. New York's move out of an uncoordinated system bundles three changes at once; Manta held coordination and digitization fixed and changed only the objective, so it isolates the one New York cannot.



Note: Both decompositions are put on a common scale, each range normalized to 100%. New York is redrawn from the span totals reported in Figure 5 of [Abdulkadiroğlu et al. \(2017\)](#) (full range 18.96 miles of willingness to travel). Their single 3.73-mile “potential algorithm improvements” band is split here, as their own printed gaps imply, into 0.62 miles of refinement among coordinated mechanisms and 3.11 miles from the ordinarily Pareto-efficient assignment, which is not itself stable, to the unconstrained optimum, so that the two columns compare like with like; Manta is Preschool 1 (full range 0.784 km of distance-equivalent welfare, from the distance rule to the unconstrained cardinal maximum—the capacity-feasible assignment of the same estimated utilities, at their posterior means, that maximizes mean welfare, an upper bound no priority-respecting mechanism can attain). The two large bands are not the same object, which is the point: New York’s move *out of* an uncoordinated system bundles coordination, digitization and acting on preferences and cannot attribute its 45.0% to any one of them, whereas Manta held coordination and digitization fixed and changed only the objective, so its 87.6% is acting on preferences alone. With the bands aligned, refinement among coordinated mechanisms is small in both—3.3% of the New York range against 0.3% of ours—and so is the residual only cardinal information could close, from the ordinarily efficient assignment to the unconstrained optimum, 16.4% against 12.1%. Robustness of the headline share to the estimation specification is reported in Figure 12 of Appendix A. Source: [Abdulkadiroğlu et al. \(2017\)](#) for New York; authors’ estimation for Manta.

[Abdulkadiroğlu et al. \(2017\)](#) find the efficiency refinements add little once offers are coordinated, we find that SIC and TTC improve almost nothing over DA where seats are slack, while *eliciting* preferences accounts for essentially the entire welfare gain. The robust cross-context message is qualitative—a small Ecuadorian city and a large U.S. metropolis agree that the algorithm matters far less than acting on preferences. The absolute magnitudes differ, as expected: [Abdulkadiroğlu et al. \(2017\)](#) study secondary-school choice, where families travel farther, whereas we study preschool and early primary, where proximity is valued more highly. A numerical comparison of the recovered welfare shares—suggestive at best, since the two studies use different welfare benchmarks—is deferred to Appendix B.1.

Survey evidence is consistent with these estimates. In an online survey administered after applications closed but before assignments were announced, families reported high confidence that they would be placed at a stated preference and high satisfaction with the application pro-

cess (Figure 8, Appendix A). These responses are not inputs to our demand model; that families expected and valued the placements the mechanism delivered is consistent with gains that reflect a system acting on preferences families already held, not a response to new information. We do not read these responses as an independent measure of the welfare gain: they were elicited under the new mechanism only, so no distance-rule counterpart exists against which to compare them.

Who gains. The average gain understates the case, because the richer heterogeneity reveals an incidence that is concentrated, and in the second entry grade progressive. It is concentrated: in Preschool 1 the 51.8% of families who gain do so by about 1.4 km each, and a quarter of those winners account for more than two-fifths of all the gains. The remainder are not, for the most part, made worse off: 31.1% of applicants are essentially unaffected (a mean gain across the lottery draws within 0.01 km of zero, so in effect the same placement under both rules), and the 17.1% who do lose give up 0.32 km on average, against the 1.43 km the winners gain. The distribution is less favourable in the later grades, where congestion binds: 33.7% lose in Preschool 2 and 57.4% in Primary 1, which is the incidence counterpart of the near-zero aggregate gain we report for that grade. And in Preschool 2 it is progressive: the gain falls with the schooling of the applicant’s neighbourhood, proxied by mean years of schooling among adults in the census block. Across the four quartiles of that proxy, from lowest to highest, the mean gain there is +0.49, +0.38, +0.34 and +0.15 km. The corresponding Preschool 1 figures are +0.79, +0.68, +0.68 and +0.59 km, but that contrast is not distinguishable from zero once resampling respects the census block, so we do not read a gradient in the entry grade. In Primary 1—a selected, mid-stream pool in a congested market whose aggregate gain is near zero—the comparison is uninformative rather than contrary.²⁷ Because the proxy is an ecological block average rather than a household measure—and on the Preschool 2 families for whom we observe both, it is uncorrelated with the mother’s own years of schooling—we read it as a statement about neighbourhoods, not about families, and about the *direction* of the endpoint contrast rather than its magnitude. In Preschool 2, then, preference-based assignment

²⁷Quartiles of block-level mean years of schooling, on the 978 and 790 welfare-sample applicants who match to the block file—96.0% and 94.3% of the 1,019 and 838 families in the welfare sample, about a quarter of them matched to the nearest block rather than by point-in-polygon. That sample is itself drawn from blocks with 1.25 and 1.27 more years of schooling than the applicants it excludes, so these are quartiles of a restricted range. In Preschool 2 the lowest-minus-highest quartile difference is +0.335 km and it is robust: resampling census blocks rather than families gives [+0.093, +0.614], a block-permutation placebo puts it at two-sided $p = 0.010$, it is significant at every bin count we tried, and most of it survives residualising on geography. In Preschool 1 the corresponding difference is +0.196 km, but it excludes zero only when the quartile cut-points are held fixed inside the bootstrap and families rather than blocks are resampled; re-cutting the quartiles in each resample gives [−0.010, +0.374] and a block-permutation placebo gives two-sided $p = 0.075$. In Primary 1 the gradient is +0.075 km and not distinguishable from zero ([−0.354, +0.485]). Adjacent quartiles are not separately distinguishable in any grade, so the evidence speaks to the endpoint contrast and not to a ranking of the middle groups. One convention bears directly on these numbers. A placement outside the modelled in-market set has no ranked valuation, and we assign it a positional stand-in—the estimated utility of the in-market program at the same position in the applicant’s distance-appended list, as under Table 8. Placements off the submitted list but inside the modelled set carry their own estimated utilities. The stand-in sits at or below the floor the family’s own ranking implies, and the resulting bias is largest for the families most exposed to out-of-market placement under the distance rule, who are concentrated in the lowest quartile. Under valuations consistent with the submitted rankings the entry-grade contrasts fall to roughly [−0.02, +0.08] and [−0.04, +0.08], both covering zero, so the gradient we report is conditional on this convention. The aggregate welfare comparison is not: the share of the attainable range that deferred acceptance recovers is 99.4–99.6% under alternative valuations of those out-of-market placements, against the 99.6% we report.

is a *concentrated* reallocation that also tilts toward less-advantaged neighbourhoods, an equity dimension the headline average conceals and that only the family-level heterogeneity makes visible. In the entry grade it is concentrated, but we cannot show that it is tilted.

Improvements over the DA algorithm are context-dependent: the scope for post-DA efficiency refinement (stable improvement cycles, top trading cycles) is greater in the congested grades, where the stability constraints deferred acceptance respects bind hardest.

6.4 What these estimates do and do not deliver

The welfare comparison rests on a small number of assumptions and valuation conventions that we have used at several points above, and we set them out here in one place: what each delivers, the direction in which each can mislead, and the evidence that disciplines it.

Placements outside the submitted list. Identification for short-list families rests on the truncation that a listed school is preferred to every unranked alternative, including nearer ones. That truncation reads an omission as a statement of preference. Our companion survey indicates that it is sometimes instead a statement of awareness: shown up to five nearby public schools they did not apply to, families report knowing only 7.7% of them well and a further 21.6% by name. The distinction matters here because the distance rule assigns without regard to what a family knows, and it places only 50% of applicants in a school they listed; wherever it assigns a family to an unlisted school, our estimates value that placement below the family's last listed school by construction. If such schools are unknown rather than unwanted, our estimates misvalue that placement: the truncation mechanically depresses its modelled value relative to treating the omission as uninformative. Whether that overstates the welfare forgone relative to the family's *true* valuation depends on how awareness correlates with desirability, which we do not observe. We do not claim the net bias is signed: an unknown school's latent value could lie above or below the bound the truncation imposes, and limited awareness may equally have kept attractive options off lists under deferred acceptance. What the argument delivers is the direction on this particular margin, not a bound on the overall gap. Two considerations bound this concern without dissolving it. First, the comparison of assignment *outcomes* does not depend on the truncation at all: the share of applicants placed in a school they themselves named rises from 50% to 78%, and in their stated first choice from 42% to 70%, and a family that named a school has affirmatively revealed that it wants it. Second, when the distance rule places a family outside its list it is generally because the school the family did name—for most single-list families, their nearest—was full, so the substitute lies *farther* away; an unknown alternative would have to carry enough unobserved desirability to overcome a distance penalty the family has already revealed it dislikes. What we cannot do with these data is price the remaining margin: helping families discover options they do not yet know is a distinct policy lever from acting on the preferences they already hold, and only the second is evaluated here.

Truthful reporting. We take the submitted lists as truthful. In the configuration we analyse, with static sibling priority, a single quota per program and school-specific tie-breaking, submitting one's true ranking is weakly dominant (Proposition 1), the pilot's communication em-

phasized that property, and applicants faced no cap on the length of their lists, which removes the truncation margin responsible for most documented departures from truth-telling in systems with short lists. The deployed platform additionally offered dynamic sibling priority and family linking, which can break that guarantee for the applicants they touch; our simulations disable both, the applicants concerned are those with a sibling applying in the same round, a small minority, and dropping the sibling term together with the sibling-linked applicants leaves the estimates and the welfare conclusions unchanged (Appendix B.1). Re-estimating demand by the stability approach of Fack et al. (2019), which sets the submitted ranking aside and requires only that each applicant weakly prefer her assignment to the schools she could ex post have been admitted to, leaves the entry-grade welfare conclusion unchanged; what it does not reach is the separate assumption that omitted schools were known and less preferred, which is the margin discussed above. On the status-quo side, the distance rule conditions on reported residence and is in principle manipulable through it, as any address-based assignment is, and we bound that channel instead of estimating it: even under heavy misreporting the access gap does not close (Section 6.1, Table 21).

Out-of-market placements and the welfare sample. Where a mechanism places an applicant in a program outside the modelled in-market set, the placement has no ranked valuation, and we assign it a positional stand-in, the estimated utility of the in-market program at the same position in the applicant’s distance-appended list; in 84–99% of such placements that program is another shift (*jornada*) of a school that is in the modelled set. The stand-in sits at or below the floor the family’s own ranking implies, and the bias it introduces is largest for the families most exposed to out-of-market placement under the distance rule, so the neighbourhood gradient we report for Preschool 2 is conditional on this convention while the aggregate comparison is not: in the entry grade the recovered share is 99.4–99.6% under alternative valuations of those placements, against the 99.6% we report. The welfare comparison is also computed on a narrower population than the assignment shares. Of the 3,984 applicants, 2,372 rank only schools inside the modelled set and 2,200 receive a comparable placement in at least one lottery draw (Table 2); the 172 applicants lost at the second step are not missing at random, since the shortfall is proportionally largest in the most congested grade and the applicants it omits place worse under either rule. We therefore read the access gain as a statement about all applicants and the welfare gain as one about the applicants whose preferences we can estimate *and* who receive a comparable placement in the simulations.

What we observe of schools. Administrative records describe these schools with little beyond a handful of facility indicators, and the imagery attributes and image embeddings we add to them are generated regressors entered into the hierarchical prior: the reported intervals do not propagate the uncertainty in deriving them from images, so we read individual coefficients as indicative and not as definitive. What the richer measures recover is the non-distance *level* of desirability that a proximity benchmark misses, and not the way families substitute across schools, since per-family coefficients on the leading imagery indices are matched by indices with no imagery content at all, and only the first index in the preschool grades clears that placebo. Most of the dispersion in ξ_j is unexplained in sample even with every observable we assemble, and school by school essentially all of it is, so school desirability enters our estimates through the school effect and not through the covariates.

The cardinal scale. Our welfare statements are cardinal, and an ordinal, strategy-proof mechanism need not raise cardinal welfare relative to a manipulable one (Abdulkadiroğlu et al., 2015), so the estimates rest on the utility scale the demand model recovers, fixed by normalizing the average disutility of a kilometre to -1 . Two things discipline it. The welfare conclusions are stable across the alternative specifications reported in Appendix B.1 and Figure 12, including the concave cost that best predicts the submitted rankings. And the scale is willingness to travel in kilometres, the currency the status quo itself acted in, within which the additional travel deferred acceptance induces is already netted out. The survey evidence can corroborate but not settle the point: it measures pre-result beliefs and process satisfaction under deferred acceptance only, with no distance-rule counterpart and no elicitation of school–distance trade-offs, and it is no input to the model.

These are statements about what the comparison is measured on: two rules re-cleared over the same applicants, the same capacities and the same lottery draws, valued in kilometres of willingness to travel, with the headline resting on the entry grade, where the conventions above bear least.

7 Discussion

Both developed and developing countries are increasingly adopting centralized choice and assignment platforms for schools (Aguilera et al., 2022; World Bank, 2024). While there is evidence on the benefits of incorporating family preferences and coordinating assignment in New York City (Abdulkadiroğlu et al., 2017), evidence from developing-country contexts remains limited. In this paper, we study the welfare effects of changing the way school seats were allocated in Manta, Ecuador. The exercise re-clears the observed market under each rule, holding the applicant pool, their estimated preferences, their locations and school capacities fixed; it therefore does not capture reform-induced changes in participation, information acquisition, residential reporting or school supply. Before the policy change, the system ignored preferences and assigned students solely by the linear distance between their declared home and schools. We evaluate the impact of adopting a system that collects rank-ordered lists and uses a deferred acceptance (DA) algorithm that is strategy-proof in the configuration we analyse (static priorities, with dynamic sibling priority and family linking disabled). We follow the empirical strategy used in Abdulkadiroğlu et al. (2017) for New York City and estimate preferences to quantify the welfare effects of the policy change and compare outcomes under different counterfactuals.

Our main result is that switching the objective of an already-centralized, algorithmic assignment system—from minimizing distance to respecting elicited preferences—yields substantial welfare benefits, holding fixed the coordination of supply and demand and the largely digital application process. Coordinating supply and demand could itself have been the source of these gains; because Manta’s prior system already coordinated and computerized assignment, we show it was not. Our estimates echo the New York findings of Abdulkadiroğlu et al. (2017) in a very different setting: the choice among coordinated algorithms matters far less than whether the system acts on preferences at all. The two exercises, however, recover different objects. New York’s reform moved out of an *uncoordinated* system, so its estimated gain bundles the value of coordination together with the value of acting on preferences and cannot separate

them; ours holds coordination fixed and identifies the second alone. Our developing-country, early-grade setting therefore complements that evidence rather than repeating it: the qualitative lesson carries across two markets that could hardly differ more, while the welfare object we estimate is one that New York’s design cannot isolate.

Specifically we find that the new allocation based on the DA algorithm increases the percentage of applicants assigned to a preferred school from 50% to 78% and first-choice assignments from 42% to 70%, with an average increase of 0.32 km in home-to-school distance. This added travel is a cost, but a small one relative to the welfare gain: in welfare-equivalent kilometres, deferred acceptance forgoes only 0.003 km against the constrained-efficient benchmark in the entry grade, where the distance mechanism forgoes 0.689 km—a gap of more than two orders of magnitude. Entry-grade applicants obtain that gain with an average increase of 0.33 km in home-to-school distance. The gain that remains for a preference-based rule to deliver is far smaller in the congested grades. The direct DC→DA gain shows this most clearly, since the efficiency benchmark cancels out of the difference: 0.686 km per applicant in Preschool 1, 0.354 km in Preschool 2 and 0.012 km in Primary 1. All three are positive, so congestion flattens what a preference-based rule can realize without disturbing the ordering of the two mechanisms. The same ordering holds as a share of the attainable range, and it is precisely estimated: deferred acceptance closes 99.6% of that range in the entry grade and 82.3% in Preschool 2, on disjoint bootstrap intervals ([99.5, 99.8] against [77.2, 86.0]; Section 6.3). In Primary 1 the gain nearly vanishes and the share is a ratio of two congestion-compressed quantities, so we read it as near zero (Appendix B.1). The gains are larger still for applicants whose assignment differs across the two mechanisms.

These welfare estimates are necessarily model-based: because each family experiences a single mechanism, the equilibrium effect of an alternative rule cannot be observed directly and must be recovered from estimated preferences—the standard structural approach to policy questions that cannot be randomized. They are also partial-equilibrium in the supply of schooling: we hold school locations and capacities fixed and re-clear the same market under each rule. A complementary literature shows that the supply side of education markets responds to choice and funding policy (Dinerstein & Smith, 2021; Singleton, 2019); Dinerstein et al. (2020) characterizes this equilibrium margin directly. They are disciplined by a hard descriptive fact, not imposed: more than a quarter of families rank a school other than their nearest first (Figure 2)—a stated choice no proximity-only model can rationalize, and survey evidence on beliefs and satisfaction is consistent with the estimated welfare ordering. What travels across settings is not a single number but a mechanism: the value of acting on preferences is largest where seats are slack and shrinks toward zero as congestion binds (the recovered welfare share falls across our three grades; Appendix B.1)—so the largest gains should arise in systems combining spare capacity, dispersed preferences, and a status quo that ignores preferences, and the smallest where seats are scarce. For practitioners this suggests a sequence: adopt a coordinated, preference-based system first, where the bulk of the welfare lies, and only then use the elicited preferences to weigh efficiency refinements such as stable improvement cycles or top trading cycles, whose scope is largest in the congested grades. The reform’s value comes neither from digitization—the system was already digital and centralized—nor from helping families locate schools: the platform displayed the feasible alternatives to every applicant, and those who ranked few schools overwhelmingly report that too few acceptable options lie close by rather than that they could not find more (Section 4). It comes instead from a centralized algorithm

finally acting on the preferences families already held. What we do not measure is the value of the options families have not yet discovered: knowledge of unlisted nearby schools is thin (Section 6.4), and closing that gap is a distinct lever from the one evaluated here.

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Data Availability Statement

A replication package containing the code that generates all estimates, simulations, tables, and figures in this paper is available to the editor and referees on request, and will be deposited in the journal’s replication archive upon acceptance. The underlying application micro-data were provided by Ecuador’s Ministry of Education under a confidentiality agreement and contain personally identifiable information; they cannot be made publicly available. Researchers seeking access should contact the authors, who will facilitate requests to the data provider. The parental survey collected by the authors is subject to the same confidentiality constraints; its instrument and tabulation code are part of the replication package, and a de-identified extract will be deposited to the extent compatible with respondent confidentiality. School-level attributes derived from satellite and street-level imagery enter as constructed covariates and are included in the package. The proprietary Street View and satellite tiles from which several of them are measured cannot be redistributed under the providers’ terms, and are not; the open-licensed and author-owned imagery can be, and a sample of it—the authors’ own survey photographs with faces blurred, Mapillary street-level images (CC BY-SA 4.0) and Copernicus Sentinel-2 chips—is published in an online appendix accompanying the paper.

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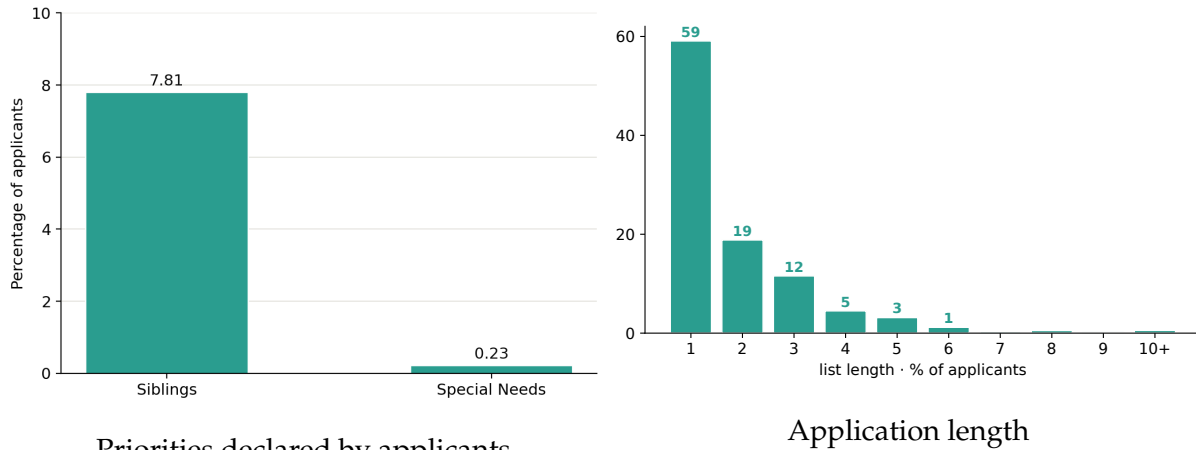
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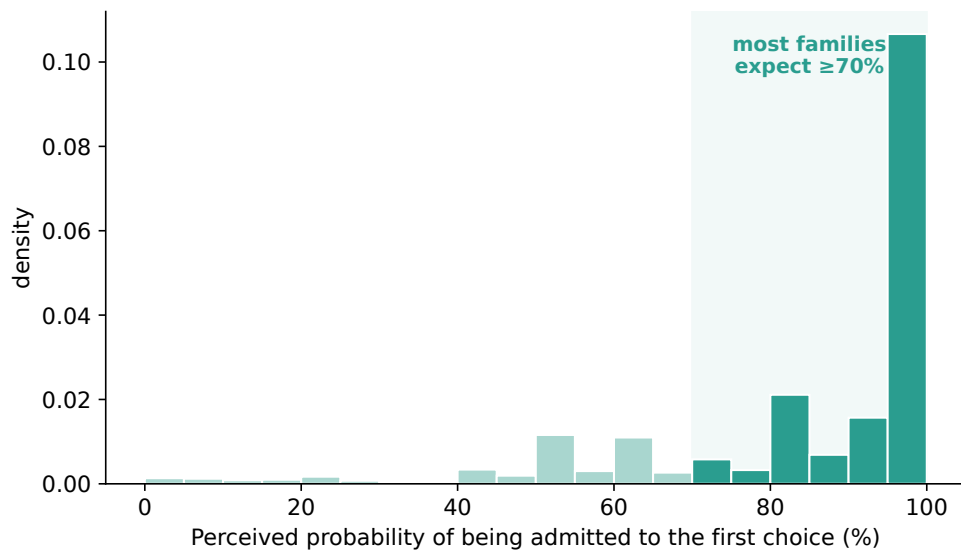
A Figures

Figure 7: Distribution of Declared Applicant Priorities and Rank-Ordered List Size



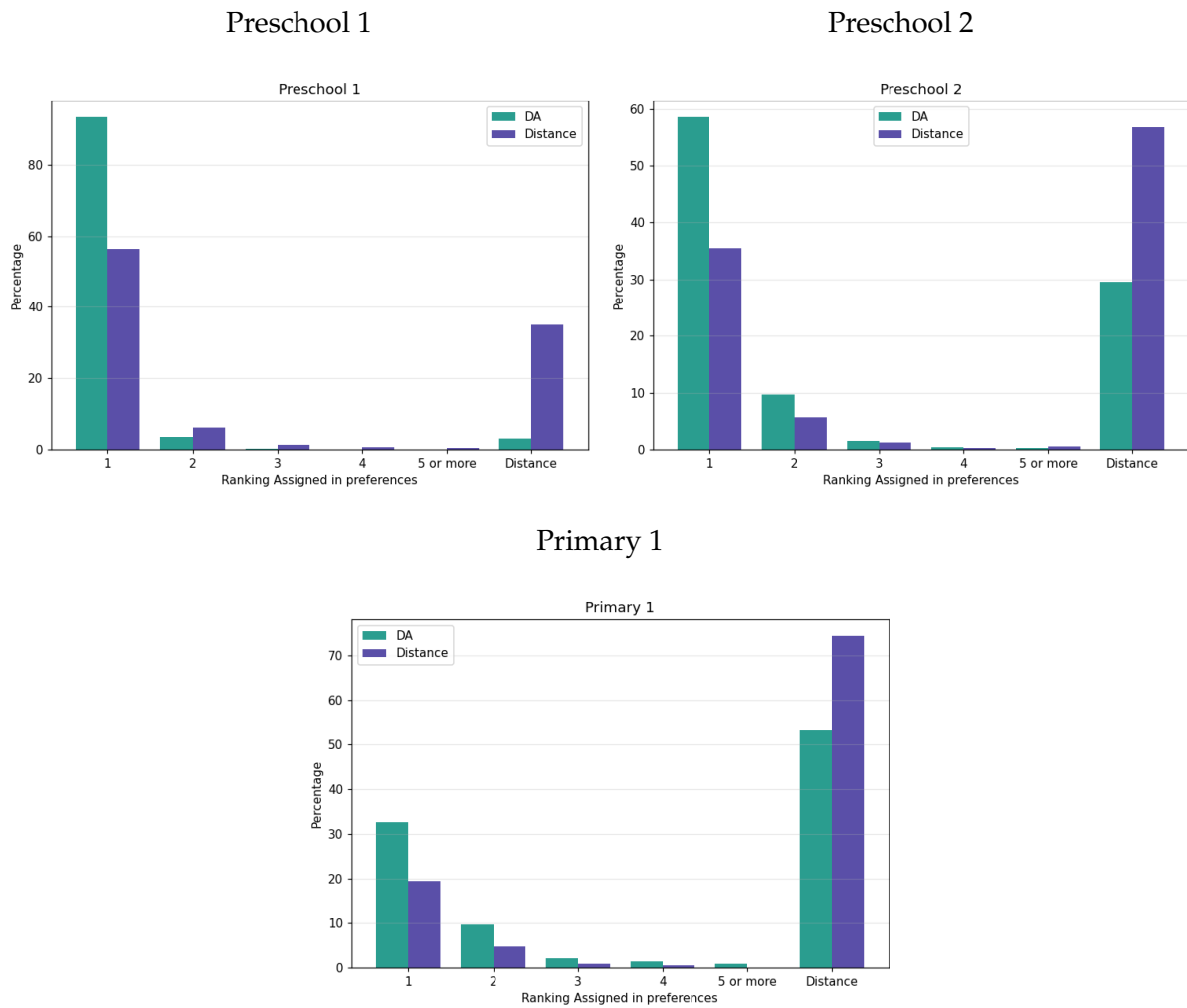
Note: Left: distribution of declared priority categories among applicants. Right: distribution of submitted rank-ordered-list lengths (percent of applicants). Pooled across grades. Source: authors' calculations from the 2021 application data.

Figure 8: Perceived Probability of Admission to 1st Preference



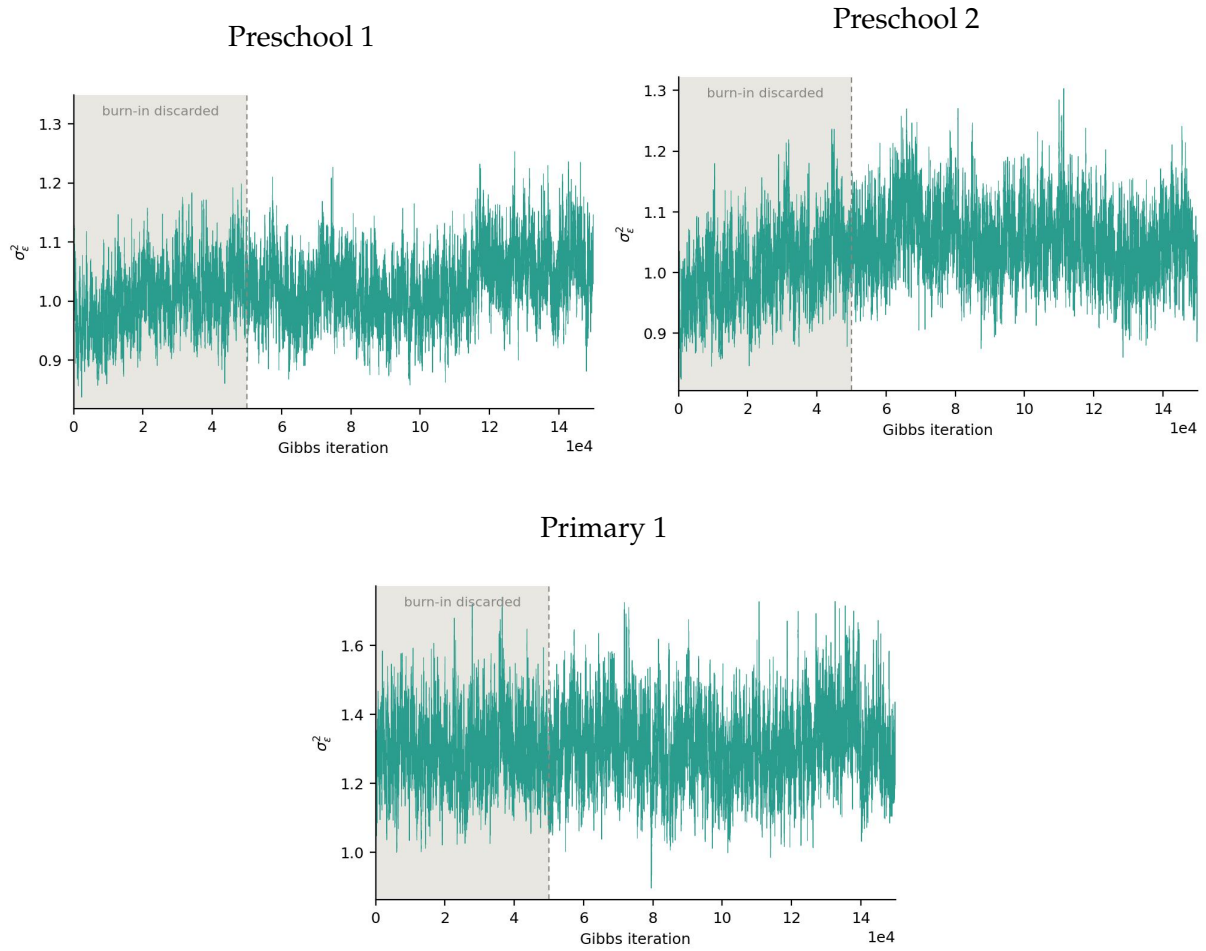
These responses were obtained in an online survey carried out after the end of the application period but before assignment results were communicated (to avoid biasing responses).

Figure 9: Ranking Assigned by Grade: DA and Distance Mechanism



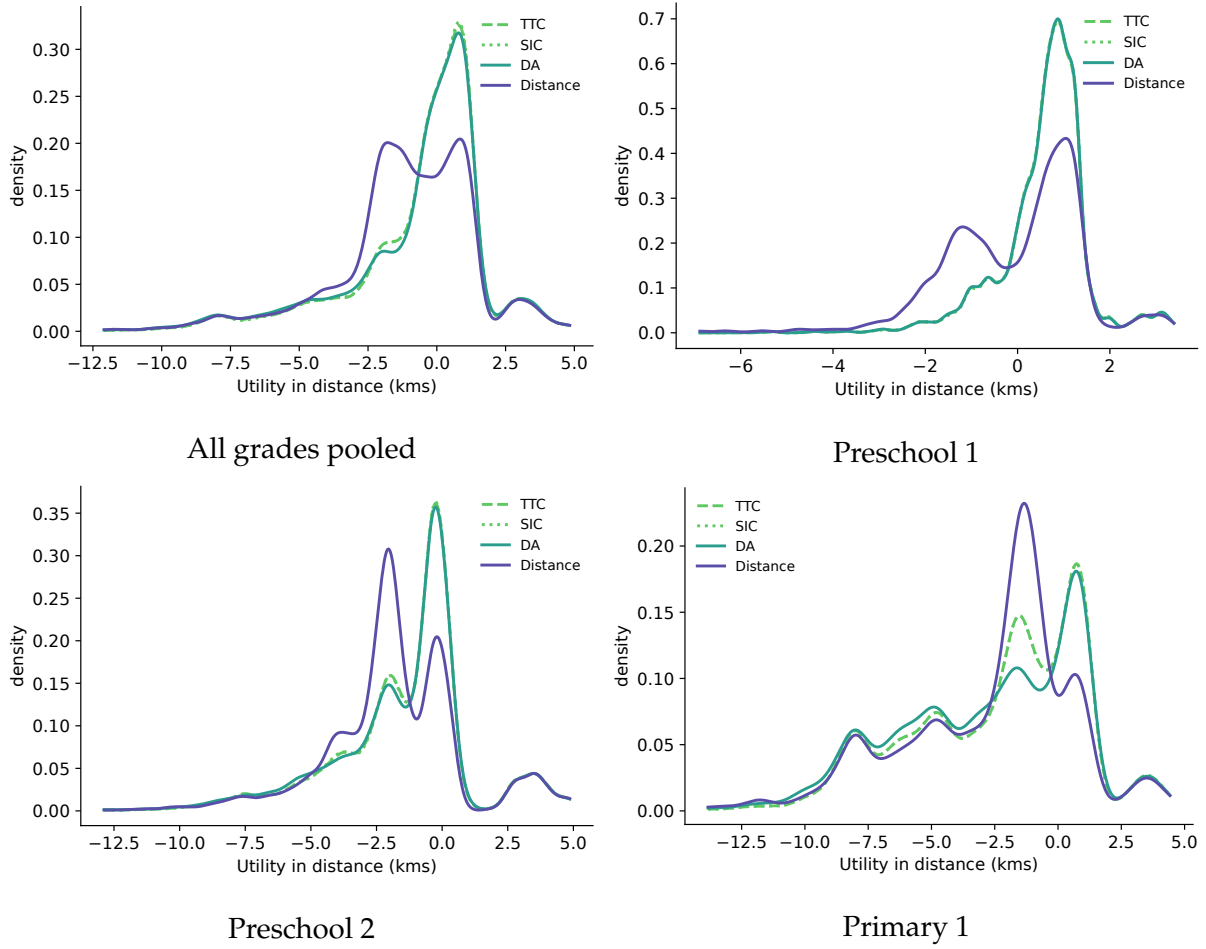
Note: Rank of the assigned school within the submitted list, by grade, under deferred acceptance and the distance rule. Source: authors' calculations.

Figure 10: Trace Plots of σ_ϵ^2 in Main Specification



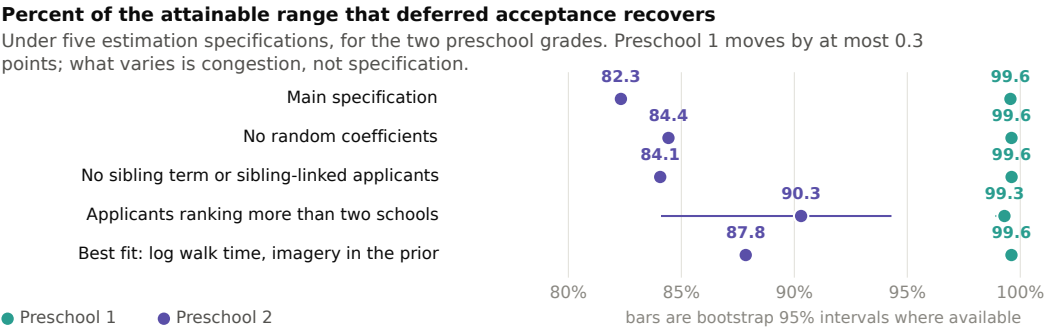
Note: Trace plots of the idiosyncratic-taste variance σ_ϵ^2 —the quantity the sampler draws, whose square root is reported in Table 10—across Gibbs iterations, by grade. Stationary traces indicate convergence for the parameter shown; convergence of the full parameter vector is assessed by the potential scale reduction factors reported in Tables 10, 16 and 17. Source: authors' estimation.

Figure 11: Welfare Distribution



All four panels are drawn on the welfare-comparison sample—the 2,200 applicants whose preferences we estimate and who also receive a comparable placement, so that the per-student welfare distribution is simulated for them (Table 2)—rather than on the full pool of 3,984 applicants, who do not all have modelled utilities. In this figure, we plot the utilities obtained with our model when using the location and scale normalizations $\bar{\delta} \equiv 0$ and -1 as the average disutility from each linear km of distance between the school and the reported location of the family. The level of utility is not relevant, as it depends on the normalization. However, the mass from the utility distribution when using the distance-centric algorithm being shifted to the left is relevant, as it indicates how the relative distributions of utilities compare, and lead to the average differences presented in Table 8.

Figure 12: Recovery of the attainable range under five estimation specifications



Note: Percent of the distance-rule-to-constrained-efficient range that deferred acceptance recovers, under five estimation specifications, for the two preschool grades. Preschool 1 moves by at most 0.3 points (99.3–99.6%); Preschool 2 ranges from 82.3% to 90.3%. What moves the statistic is congestion, not the modelling choice. The fifth specification is the one that best predicts the submitted rankings out of sample: the cost is $\log(1 + \cdot)$ of walk time placed on a kilometre-equivalent scale, and the ten imagery attributes together with the eight image dimensions enter the prior on ξ_j . Because that cost is not linear in kilometres, the underlying utility levels are not denominated in kilometres; the recovered share is a ratio of utility differences and is unaffected. It is estimated with the replication package’s Gibbs sampler, which reproduces the main specification’s recovery to 0.07 points in both grades. Bars are applicant-level bootstrap 95% intervals where available. Source: authors’ estimation.

B Tables

Table 9: Comparative Statistics for Guayaquil and Manta

	Guayaquil	Manta
Total population	2,291,158	221,122
Population 3-5 years old (% of population 3-17 years old)	19.7	19.1
Minors in the school system (% of population 3-17 years old)	78.9	80.2
Average Mother's Education (of minors 3-17 years old)	11.3 years	10.8 years
Total schools	885	153
Share of public schools	54%	43%
Share of private schools	44%	54%
Share of "fiscomisional" schools	2%	3%
Total enrolment	687,046	86,455
Share of enrolment in public schools	57%	67%
Share of enrolment in private schools	40%	25%
Share of enrolment in "fiscomisional" schools	4%	8%

Note: Comparative statistics for Guayaquil and Manta, from the 2010 Census and school-transfer requests in the 2019–2020 school year. Source: authors' tabulation.

Table 10: Estimates and Potential Scale Reduction Factors: Main Specification

	Preschool 1		Preschool 2		Primary 1	
Estimate	Mean (SD)	PSRF	Mean (SD)	PSRF	Mean (SD)	PSRF
ξ_j	-0.035 (0.485)		0.001 (0.501)		-0.006 (0.628)	
λ	3.043 (0.360)	1.001	4.507 (0.557)	1.003	3.789 (0.824)	1.000
σ_{ξ}	0.529 (0.058)	1.001	0.555 (0.065)	1.001	0.714 (0.092)	1.000
σ_{ϵ}	1.016 (0.027)	1.008	1.023 (0.028)	1.006	1.145 (0.045)	1.002
σ_{γ}	1.225	1.018	1.540	1.056	1.111	1.001
Tot. schools	55		57		54	
Tot. students	1,098		885		389	

Note: Posterior means of the random-utility model estimated by Gibbs sampling, with potential scale reduction factors (Gelman et al., 1992) (values near one indicate convergence). Utility is in kilometres, with the average disutility of distance normalized to -1 . The dispersion parameters σ_{ξ} , σ_{ϵ} and σ_{γ} are reported as standard deviations. For σ_{ξ} and σ_{ϵ} the sampler draws the corresponding variances, and the posterior standard deviations in parentheses follow by the delta method. No parenthesis appears in the σ_{γ} row: the retained posterior summary for the distance-taste variance stores its mean and its scale reduction factor alone, so the table reports those two for it. How σ_{γ} should be read depends less on its posterior precision than on its content: Section 5.3 shows that a third to a half of it is curvature in the cost function rather than dispersion in tastes, so that it is an upper bound on taste heterogeneity. σ_{ξ} and σ_{ϵ} are denominated in kilometres and so are directly comparable with λ and ξ_j . σ_{γ} is not: it is the dispersion of a multiplicative shifter on the distance coefficient and is therefore dimensionless.^a Scale reduction factors are computed on the variance chains. The additive level of ξ is not identified by a rank-ordered likelihood—adding a constant to every ξ_j leaves all rankings unchanged—so we normalize ξ to have mean zero across schools within each posterior draw and report σ_{ξ} as dispersion about that grand mean. The ξ_j row accordingly reports the cross-school *mean* of the posterior-mean school effects—near zero by construction, though not identically so, since it is an average of draw-wise centred vectors—with the cross-school standard deviation in parentheses. That parenthetical is a dispersion across schools, not a posterior standard deviation, which is why it tracks σ_{ξ} rather than the small posterior uncertainties reported for the other rows. Source: authors’ estimation.

^aBecause the average disutility of distance is normalized to -1 , a family’s effective distance coefficient is $-1 + \gamma_i$, so a value $\gamma_i > 1$ would describe a family that prefers schools farther from home. The population dispersion is not the realized distribution: γ_i is estimated only for the families who rank more than two schools and is held at the population value for the rest, and the estimated family-specific coefficients are shrunk accordingly. In Preschool 1, integrating each family’s own posterior, the share with a positive marginal utility of distance is 0.25% across all applicants and 1.19% among the families for whom γ_i is estimated; the posterior mean of γ_i is -0.11 . Applying the population standard deviation as though it described individual families would overstate that share by roughly two orders of magnitude.

Table 11: Mechanism Comparison - Results Preschool 1

	DA	DC
Panel A: Applicants assigned in:		
<i>Any preference</i>	1,643 (96.31%)	1,105 (64.80%)
<i>First preference</i>	1,592 (93.31%)	963 (56.45%)
<i>Average assignment distance</i>	0.85 km	0.52 km
Panel B: Applicants with the same 1st preference (982) assigned in:		
<i>Any preference</i>	955 (97.26%)	956 (97.38%)
<i>First preference</i>	936 (95.31%)	949 (96.63%)

Note: Assignment outcomes under deferred acceptance and the distance rule for Preschool 1, mean over 100 lottery draws (fixed seed). Population shares in parentheses. Source: authors' calculations.

Table 12: Mechanism Comparison - Results Preschool 2

	DA	DC
Panel A: Applicants assigned in:		
<i>Any preference</i>	1,201 (70.57%)	736 (43.23%)
<i>First preference</i>	992 (58.30%)	605 (35.52%)
<i>Average assignment distance</i>	1.37 km	1.08 km
Panel B: Applicants with the same 1st preference (960) assigned in:		
<i>Any preference</i>	690 (71.91%)	602 (62.67%)
<i>First preference</i>	608 (63.35%)	566 (58.98%)

Note: Assignment outcomes under deferred acceptance and the distance rule for Preschool 2, mean over 100 lottery draws (fixed seed). Population shares in parentheses. Source: authors' calculations.

Table 13: Mechanism Comparison - Results Primary 1

	DA	DC
<i>Panel A: Applicants assigned in:</i>		
<i>Any preference</i>	265 (46.81%)	149 (26.29%)
<i>First preference</i>	186 (32.72%)	115 (20.23%)
<i>Average assignment distance</i>	2.66 km	2.29 km
<i>Panel B: Applicants with the same 1st preference (283) assigned in:</i>		
<i>Any preference</i>	144 (50.71%)	116 (41.07%)
<i>First preference</i>	113 (39.85%)	106 (37.53%)

Note: Assignment outcomes under deferred acceptance and the distance rule for Primary 1, mean over 100 lottery draws (fixed seed). Population shares in parentheses. Source: authors' calculations.

Table 14: Assignment Inside and Outside Reported Preferences under the DA and DC Algorithms

	Preschool 1	Preschool 2	Primary 1
Applicants assigned to same schools under DA and DC	958 (56.15%)	789 (46.36%)	263 (46.38%)
Applicants assigned to different schools under DA and DC	748 (43.85%)	913 (53.64%)	304 (53.62%)
<i>Applicants assigned to same schools under DA and DC</i>			
Both DA and DC in preferences	933 (97.39%)	622 (78.83%)	114 (43.35%)
Both DA and DC out of preferences	25 (2.61%)	167 (21.17%)	149 (56.65%)
<i>Applicants assigned to different schools under DA and DC</i>			
DA in preferences and DC out of preferences	563 (75.27%)	485 (53.12%)	125 (41.12%)
DA out of preferences and DC in preferences	17 (2.27%)	21 (2.30%)	5 (1.64%)
Both DA and DC in preferences	158 (21.12%)	92 (10.08%)	26 (8.55%)
Both DA and DC out of preferences	10 (1.34%)	315 (34.50%)	148 (48.68%)

Note: Applicants cross-classified by whether they are assigned to a reported preference under deferred acceptance and under the distance rule, by grade; single representative lottery draw. Source: authors' calculations.

B.1 Appendix Robustness Checks

Welfare in the later-entry grades and the congestion gradient. We headline Preschool 1 because it is the earliest point of entry and the least congested grade, so its applicants are the least selected of the three—though not unselected: enrolment at this level is not compulsory, and Preschool 2 draws about as many applicants as Preschool 1 (Table 1), so a substantial part of the eventual cohort enters later. Preschool 1 applicants are therefore selected on early-enrolment demand, and a further 40% do not reach the welfare sample—most (611 of 1,709) because they ranked private, *fiscomisional* or out-of-market programs and so fall outside the demand model, plus 79 more without a comparable placement in the simulations; the two margins do not push the same way, and the revealed-taste argument that follows applies to the first, larger group only. Early enrolment plausibly selects on stronger non-proximity preferences, which would raise the gain in this grade relative to later entrants; but excluding families who ranked out-of-market programs removes applicants who had already revealed a taste for schools beyond their neighbourhood, which would lower the gain in the retained sample relative to all Preschool 1 applicants. We have no evidence on the relative size of the two, so we do not sign the net direction. Applicants to Preschool 2 and Primary 1 are instead a selected, mid-stream population:

most children roll over to the next level automatically, so those who *apply* at these grades are disproportionately families that moved, left a school, or were otherwise disrupted—a group whose preferences are harder to model and interpret. With that caveat, the share of the attainable welfare range that deferred acceptance recovers is far smaller in the congested grades: 99.6% in Preschool 1, 82.3% in Preschool 2, and only 3.9% in Primary 1. The gradient is itself informative—it moves with congestion as an economic channel predicts, though the applicant pools also differ across grades, so we read it as consistent with the congestion primitive rather than as isolating it—but the Primary 1 figure is imprecisely identified. The Preschool 2 figure is insensitive to how the school effects are normalized: re-valuing the same simulated assignments with utilities estimated under the mean-zero normalization of ζ (Table 10) leaves the recovered share at 82.4%. An earlier reconstruction that re-simulated the assignments on the in-market pool alone suggested a substantially higher share, but that restriction drops—non-randomly—the applicants the distance rule serves worst, precisely because those applicants are assigned outside the estimated choice set; it therefore understates the cost of the distance rule and overstates the share deferred acceptance recovers. The Primary 1 figure is a ratio of two congestion-compressed quantities (a bootstrap interval of roughly $[-77, 36]\%$) and is best read as “near zero.” Finally, the Preschool 2 share is in the same range as the roughly 80% that [Abdulkadiroğlu et al. \(2017\)](#) report for the New York City high-school match; we read this numerical agreement as suggestive rather than a matched magnitude, since the two ratios use different welfare benchmarks—New York’s range runs from an *uncoordinated* baseline and so embeds the value of coordination, whereas ours holds coordination fixed.

Robustness to truth-telling: stability-based estimation. The preference estimates treat each submitted rank-ordered list as truthful—justified by the strategy-proofness of the static-priority deferred-acceptance configuration (Section 5), but a property the prevalence of short lists could strain. We therefore re-estimate demand following the stability approach of [Fack et al. \(2019\)](#), which does not use the reported ranking at all: it assumes only that the realized matching is *stable*, so that each applicant weakly prefers her assigned school to every school in her ex-post *feasible* set—the programs whose realized admission cutoff her priority-and-lottery score cleared, with undersubscribed programs feasible to everyone in the grade. Because this uses only the realized assignment and ex-post feasibility, it is robust to misreporting *within* the submitted ranking, which a truth-telling estimator would mishandle. Its reach is not unlimited: it substitutes the assumption that the realized matching is stable under *true* preferences, which it cannot itself verify. If a family omitted a feasible school it genuinely preferred to its assignment—from limited awareness rather than strategy—that omission violates the maintained stability condition too. We keep the identical utility specification and recompute the welfare comparison on the same mechanisms and lottery draws, changing only the demand valuation. In the entry grade, on which our headline rests, the conclusion is unchanged: deferred acceptance recovers 99.8% of the distance-rule-to-frontier welfare range under the stability estimator, against 99.9% under the baseline rank-ordered estimator. Both figures are computed on a common in-market basis, and neither is the 99.6% headline: as noted above, the in-market restriction drops the applicants the distance rule serves worst and so overstates the recovered share. The comparison to draw is between the two estimators on that shared basis, not between either of them and the headline.²⁸ The structural parameters shift under the weaker identification—the school-effect

²⁸We also flag the limits of this exercise. In a grade where deferred acceptance places 93% of applicants in their reported first choice and most programs are undersubscribed, an applicant’s ex-post feasible set is close to the

dispersion σ_{ξ} in particular is less tightly pinned—but the welfare object we report is invariant to dropping truth-telling. As with the welfare levels above, the stability estimator is only weakly identified in the selected later grades—a larger share of their applicants are assigned outside the modelled in-market set and oversubscription is thin—where it returns implausibly large sibling coefficients; we therefore do not lean on its later-grade point estimates, consistent with basing the headline on the entry grade.

Table 15: Willingness to Travel for School-Plant Scale (imagery)

WTT (km per +1 SD)	Preschool 1	Preschool 2	Primary 1
Building footprint	+0.42*	+0.36*	+0.43*
Building count	−0.25*	−0.33*	−0.42*

Note: Posterior means from the random-coefficient specification with a hierarchical prior $\xi_j \sim N(X_j\beta, \sigma_{\xi}^2)$ on the ten standardized imagery attributes listed in the text; entries are willingness to travel (km) for a one-standard-deviation increase. * 95% credible interval excludes zero. Building footprint and count are the only two of the ten attributes significant in every grade; the remaining eight ground-quality attributes are individually imprecise and are omitted here for brevity.

Table 16: Estimates and Potential Scale Reduction Factors. Main Specification without Random Coefficients

Estimate	Preschool 1		Preschool 2		Primary 1	
	Mean (SD)	PSRF	Mean (SD)	PSRF	Mean (SD)	PSRF
ξ_j	−0.004 (0.531)		−0.042 (0.614)		−0.002 (0.594)	
λ	3.505 (0.402)	1	4.607 (0.539)	1	4.330 (0.932)	1
σ_{ξ}	0.574 (0.062)	1.000	0.666 (0.082)	1	0.688 (0.087)	1
σ_{ϵ}	1.186 (0.023)	1	1.150 (0.027)	1	1.305 (0.045)	1
Tot. schools	55		57		54	
Tot. students	1,098		885		389	

Note: Posterior means and potential scale reduction factors for the specification without random coefficients, by grade. Dispersion parameters are reported as standard deviations in kilometres, as in Table 10. Source: authors' estimation.

whole market, so the stability restriction—that the assignment is weakly preferred to the schools she could have been admitted to—nearly coincides with the restriction that her reported first choice is preferred to the market. The exercise therefore tests robustness to strategic *ordering* of a submitted list, but has little power against the distinct concern of Artemov et al. (2023) and of limited awareness (Section 6.4): schools omitted from the list altogether. Both estimators impose that listed schools are preferred to unlisted ones.

Table 17: Estimates and Potential Scale Reduction Factors. Main Specification without Siblings

	Preschool 1		Preschool 2		Primary 1	
Estimate	Mean (SD)	PSRF	Mean (SD)	PSRF	Mean (SD)	PSRF
$\bar{\zeta}_j$	-0.024 (0.491)		0.001 (0.498)		0.000 (0.625)	
σ_{ζ}	0.534 (0.060)	1.000	0.550 (0.064)	1	0.714 (0.093)	1
σ_{ϵ}	1.034 (0.025)	1.000	0.992 (0.027)	1.001	1.148 (0.046)	1.000
σ_{γ}	1.273	1.001	1.482	1.000	0.970	1.002
Tot. schools	55		57		54	
Tot. students	1,021		839		345	

Note: Posterior means and potential scale reduction factors for the no-sibling specification, by grade. This exercise changes the estimation sample as well as the specification: it drops the sibling term *and* is estimated on the applicants without a sibling link, so the counts fall from 1,098/885/389 in Table 10 to 1,021/839/345 here. It is therefore a joint check that neither the sibling term nor the sibling-linked applicants drive the results, not an isolation of the term alone. The Preschool 2 and Primary 1 columns are estimated from longer chains (400,000 draws, 100,000 burn-in) than the common length used elsewhere, because σ_{γ} —identified off the applicants who rank more than two schools, of whom there are fewest here—had not converged at the shorter length. Dispersion parameters σ_{ζ} and σ_{ϵ} are reported as standard deviations in kilometres and σ_{γ} is dimensionless, as in Table 10. Source: authors' estimation.

Table 18: Differences in Welfare: Constrained-Efficient (TTC) Benchmark vs. DC and DA Algorithms. Specification without Random Coefficients

	Preschool 1		Preschool 2		Primary 1	
Measure	DC	DA	DC	DA	DC	DA
Panel A: All simulated applicants						
Δ Mean utility (km)	-0.773	-0.003	-0.456	-0.071	-0.317	-0.296
$\frac{\Delta \text{Mean utility}}{\sigma_{\text{Ut. TTC}}}$	-0.750	-0.003	-0.177	-0.027	-0.091	-0.085
Panel B: Applicants with different assignments across algorithms						
Δ Mean utility (km)	-1.667	-0.006	-0.707	-0.109	-0.415	-0.388
$\frac{\Delta \text{Mean utility}}{\sigma_{\text{Ut. TTC}}}$	-1.631	-0.006	-0.302	-0.047	-0.124	-0.116

Note: Δ Mean utility (km) is measured computing $u_{i,j(\mu)} - u_{i,j(\text{TTC})}$, where $j(\mu)$ represents the school to which individual i is assigned under mechanism μ . We then compute average utilities for each algorithm and simulation and finally compute the average for each algorithm across simulations. $\frac{\Delta \text{Mean utility}}{\sigma_{\text{Ut. TTC}}}$ simply uses the standard deviation of utility under the TTC mechanism to scale this difference in each simulation. This is done to facilitate extrapolations to other contexts.

Table 19: Differences in Welfare: Constrained-Efficient (TTC) Benchmark vs. DC and DA Algorithms. Specification without Siblings

Measure	Preschool 1		Preschool 2		Primary 1	
	DC	DA	DC	DA	DC	DA
Panel A: All simulated applicants						
Δ Mean utility (km)	-0.781	-0.003	-0.433	-0.069	-0.320	-0.296
$\frac{\Delta \text{Mean utility}}{\sigma_{\text{Ut. TTC}}}$	-0.866	-0.003	-0.204	-0.033	-0.097	-0.089
Panel B: Applicants with different assignments across algorithms						
Δ Mean utility (km)	-1.684	-0.006	-0.670	-0.107	-0.419	-0.388
$\frac{\Delta \text{Mean utility}}{\sigma_{\text{Ut. TTC}}}$	-1.615	-0.006	-0.294	-0.047	-0.128	-0.118

Note: Δ Mean utility (km) is measured computing $u_{i,j(\mu)} - u_{i,j(\text{TTC})}$, where $j(\mu)$ represents the school to which individual i is assigned under mechanism μ . We then compute average utilities for each algorithm and simulation and finally compute the average for each algorithm across simulations. $\frac{\Delta \text{Mean utility}}{\sigma_{\text{Ut. TTC}}}$ simply uses the standard deviation of utility under the TTC mechanism to scale this difference in each simulation. This is done to facilitate extrapolations to other contexts.

Table 20: Priorities and Assignments in DA: Potential Improvements for SIC and TTC

	Preschool 1	Preschool 2	Primary 1
Panel A: <i>Ranking of schools where an applicant has sibling priority(*)</i>			
1st preference	100	157	40
2nd preference	7	3	1
3rd preference	1	2	0
Panel B: <i>Ranking of DA assignments for applicants with sibling priority below 1st preference</i>			
1st preference	8	3	0
2nd preference	0	2(**)	1(***)
3rd preference	0	0	0

Note: None of the potential applicants that could participate in a *sibling-priority* improvement cycle (Panel B) coincide in the programs to which they were applying, so no such cycle can be released. The table addresses the sibling-priority margin only; cycles that school-specific lottery reversals could open among regular applicants are ruled out by the direct cycle search over the full appended lists reported in Appendix E, not by this table.

(*) Panel A shows the highest ranked program where applicants have a sibling priority. If an applicant has priority in both the 1st and 2nd preference, they will only appear in the 1st preference in this table.

(**) One of these two applicants had sibling priority in their second preference, and the other had sibling priority in their third preference.

(***) This applicant had sibling priority in their second preference.

Table 21: Mechanism Comparison with Location Misreporting

	<i>Applicants assigned to any preference</i>	<i>Average Distance</i>
Distance Mech without misreporting	1,990 (50.05%)	1.01 km
Distance Mech + 10% misreporting	2,048 (51.53%)	1.04 km
Distance Mech + 30% misreporting	2,150 (54.08%)	1.09 km
Distance Mech + 50% misreporting	2,234 (56.20%)	1.15 km
Distance Mech + 70% misreporting	2,332 (58.66%)	1.22 km
Distance Mech + 90% misreporting	2,427 (61.06%)	1.28 km
DA Algorithm	3,109 (78.22%)	1.33 km

Note: Share of applicants assigned to a listed preference under the distance rule as the fraction misreporting their home address toward their most preferred school rises, against the deferred-acceptance benchmark. Misreporters at each level are a fixed set constructed once upstream and held constant across the draws; the only run-time randomness is the seeded lottery, so entries are means over lotteries, not over selections of misreporters. The exercise is a stress test under this one manipulation rule, not a bound over strategically chosen misreports. Mean over 100 lottery draws at a fixed seed; the baseline row is recomputed within this simulation's own draws and differs from the 50.07% of Table 7 only by across-draw Monte-Carlo variation. Source: authors' calculations.

C Phases of the Distance-Centric Algorithm Implementation Process

The overall process started with the Preparation Phase, in which the Ministry of Education updated all school supply information (i.e., location, available spaces, closure or opening of educational programs, etc.).

In the second, or Registration Phase, families registered their children on a website in order to be granted a spot in a public school. Legal guardians needed to indicate the type of registration (individual or sibling group), the grade level to be attended, any older siblings already enrolled in the public school system, special educational needs, and nationality. They also provided their electricity bill number so as to be geolocated.

This was followed by the Assignment Phase and then the Consultation Phase, during which time families could enter the website to see their school assignments. Finally, the fifth and sixth phases consisted of the School Change Petitions Phase and Continuous Enrolment. Applicants could ask to change schools if there were spaces available, and they could also enrol in a given school once the academic year had already started.

D Full Utility Specification

$$\begin{aligned}
u_{ij} &= S_{ij}\lambda + \delta_j - d_{ij} + \gamma_i d_{ij} + \epsilon_{ij} \\
\delta_j &= \bar{\delta} + \xi_j \\
\bar{\delta} &\equiv 0
\end{aligned}$$

$$\begin{aligned}
\lambda &\sim \mathcal{N}(0, \sigma_\lambda^2) \\
\gamma_i &\sim \mathcal{N}(0, \sigma_\gamma^2) \\
\xi_j &\sim \mathcal{N}(0, \sigma_\xi^2) \\
\epsilon_{i,j} &\sim \mathcal{N}(0, \sigma_\epsilon^2) \\
\sigma_\gamma^2 &\sim IW(\tau_\gamma, df_\gamma) \\
\sigma_\xi^2 &\sim IW(\tau_\xi, df_\xi) \\
\sigma_\epsilon^2 &\sim IW(\tau_\epsilon, df_\epsilon)
\end{aligned}$$

We follow [Rossi et al. \(1996\)](#) and [Abdulkadiroğlu et al. \(2017\)](#) in using diffuse priors. The only exception is the use of a smaller τ_γ , given that in this context it is reasonable to impose a smaller prior on the mean variance of the parameter, considering that $\gamma_i > 1$ would imply that a family actually prefers schools farther away from home. Specifically, we use $\sigma_\lambda^2 = 100$, $\tau_\gamma = 0 + size(\gamma_i) = 1$, $df_\gamma = 3 + size(\gamma_i) = 4$,²⁹ $\tau_\xi = 1$, $df_\xi = 2$, $\tau_\epsilon = 3 + n_{schools}$, and $df_\epsilon = 3 + n_{schools}$.

²⁹This implies that the mean of the σ_γ^2 prior is 0.5.

E DA-SIC and TTC Equivalence in our Context

Proposition 2 (SIC = TTC in the Manta environment). *Let μ^{DA} be the applicant-proposing deferred-acceptance matching and μ^{SIC} its stable-improvement-cycles refinement (Erdil & Ergin, 2008). In our environment, top trading cycles (Abdulkadiroğlu & Sönmez, 2003) applied to μ^{SIC} returns μ^{SIC} unchanged, so μ^{SIC} admits no further improving trade from that endowment and is Pareto-efficient for students. Moreover, the standard top-trading-cycles assignment computed from priorities and reported preferences coincides with μ^{SIC} in all 100 simulations, so $\mu^{SIC} = \mu^{TTC}$ throughout our results.³⁰*

The result follows from the structure of priorities and preferences here. Any residual improvement of TTC over SIC must come from a priority-trading cycle, which requires discordance between preferences and priorities (Erdil & Ergin, 2008): some applicant must strictly prefer a program at which she would displace a strictly higher-priority enrollee. On the reported-versus-appended margin there is no such discordance, because every reported program is both strictly preferred to, and carries strictly better priority than, the distance-imputed alternatives appended to the list. Two sources of priority heterogeneity remain *within* reported lists. The first is the static sibling priority, recorded for 311 applicants (316 applicant-program pairs) across the three grades (Panel A of Table 20). The second is the school-specific tie-breaking itself, which can reverse the relative ranking of two regular applicants across two schools and so, in principle, open a trading cycle among applicants who share no sibling priority at all—an argument from priority alignment alone does not rule this out. These applicants do not form a releasable trading cycle (Panel B): the few assigned below a sibling-priority program demand a common, fully subscribed set of programs rather than one another’s seats, so no seat can be vacated to route a trade. A direct cycle search over the full appended lists confirms there are no improving trades.

The Top Trading Cycles (TTC) algorithm includes the possibility of trading priorities between applicants, which happens when they prefer the alternatives in which they do not have the priority more than ones in which they do, and are thus “willing to trade” the priority. In other words, TTC has the potential to provide improvements over SIC when there is not a complete correlation between priorities and preferences.

To illustrate this, imagine a system with two schools (A and B), both with only one vacancy, and three applicants (i , j and k). i has priority in A but prefers B over A. j has priority in school B, but prefers A over B. k has priority in both schools, prefers A over B, and has the worst lottery number of the system. The result of the DA and SIC assignment would be i assigned to A and j assigned to B. The TTC algorithm would allow them to trade their priorities and switch their assignments. With that assignment switch, applicant k —still unassigned—now holds higher (pre-trade) priority at both schools that rejected him. Such a situation can only arise when the correlation between preference and priority is not one, thus leaving room to trade the priority and get a better assignment.

³⁰The first statement and the second are distinct: returning μ^{SIC} unchanged from the μ^{SIC} endowment establishes Pareto efficiency, not that the two mechanisms select the same Pareto-efficient allocation. The coincidence of μ^{SIC} and μ^{TTC} is an empirical finding in this environment, not a corollary, and the argument below explains why the priority structure here makes it so.

Online Appendix

The Pre-Results Household Survey

This appendix documents the household survey fielded to families participating in the Manta assignment: its design and fielding (§F), the instrument and questions (§G), coverage and representativeness (§H), detailed results (§I), and a summary of what the survey establishes (§J). All tabulations are reproducible from the survey microdata; the analysis code is included in the replication package.

F Survey design and fielding

The survey was administered to families participating in the 2021–2022 (Costa–Galápagos regime) centralized school-assignment pilot covering the cantons of **Manta**, **Jaramijó**, and **Montecristi**. It was a *pre-results* instrument: families were surveyed after submitting their ranked applications but *before* assignment results were released. This timing is deliberate—it elicits beliefs and stated reasons at the moment of application, uncontaminated by the realized outcome. Three design features are central.

(1) Personalization. The instrument was individualized for each family using embedded data from the application platform. Respondents saw their own ranked list, their child’s name, and the number of schools they had applied to, and were asked about *specific* schools they named as known. This ties answers to each family’s actual choice set.

(2) Belief elicitation with randomized framing. Subjective admission probabilities were elicited on a 0–100 scale, including hypotheticals for the second and third preferences, and a positively- vs. negatively-framed item on the probability of obtaining *at least one* (or *no*) seat. The frame was randomized across respondents.

(3) Linkage. Each response carries the applicant identifier, which matches the assignment data (99.8%) and the geocoded home file (100%), so the survey can be linked to realized assignments, home locations, and—via a census block join—neighbourhood socioeconomic status.

Ratings use Ecuador’s 0–20 school-grade scale (a “nota”), where 16–20 is the top band. Invitations were delivered through the platform’s messaging channels (SMS, WhatsApp, e-mail, and web).

G The instrument

We reproduce the analytically central items (original Spanish, with an English gloss); [child], [1st pref], etc. denote piped personalization fields. The full questionnaire (Qualtrics .qsf and a Word version) is in the replication package.

Module 1 — Application list. *Why not list more:* “¿Por qué no agregó más instituciones educativas a su postulación? (Marque la razón principal)” — “Why did you not add more schools?

(Mark the main reason)”; single choice among (a) no public school close enough, (b) confident of a seat in a listed school, (c) hard to find more schools, (d) knows the others and prefers to remain unassigned, (e) would enrol private if unassigned. *Needed more information:* “Para convencerse de agregar estas instituciones adicionales, ¿tuvo que buscar más información?” *Perceived effect of a longer list:* “Si agrega más instituciones a su postulación, ¿qué cree que pasaría?”—“lowers the chance of going unassigned” and/or “lowers the chance at my top preferences.”

Module 2 — School knowledge. The instrument displayed up to five nearby public and five private schools the family had *not* applied to, and asked how well the respondent knew each (“la conozco bien” / “sólo de nombre” / “no la conozco”). The schools were shown to the respondent; they were not named by the respondent.

Module 3 — Beliefs about admission. “Usted eligió a [1st pref] en primera preferencia para [child]: en una escala del 0 a 100, ¿con qué probabilidad cree que va a obtener un cupo en esa opción?”, and the same hypothetically placing the 2nd / 3rd choice first. A randomized frame asked the probability that [child] *will* obtain at least one seat / *will not* obtain any seat among the listed schools.

Module 4 — Satisfaction and platform ratings (0–20). Overall rating of the assignment system; ratings of school information, process information, and platform ease of use; and satisfaction if assigned to the first preference, the last listed preference, or no school.

Module 5 — Background. Mother’s/guardian’s education and information sources used.

H Coverage and representativeness

The survey drew **1,873 respondents** (roughly **47% of the ~3,984 applicants**), of whom **1,517** (81% of respondents; 38% of applicants) completed the full instrument, across all three entry grades. Figure 13 maps the binned density of surveyed families’ home locations and the schools; spatial coverage tracks the applicant population across the Manta urban core and outlying parishes.

Survey respondents are mildly positively selected but well balanced on outcome-relevant margins (Table 22): slightly more likely to be Manta-origin and to live in lower-deprivation blocks and to submit marginally longer lists, but balanced on grade, true admission risk, and realized placement, with all standardized differences ≤ 0.17 . We treat the survey as a credible validation sample (with light reweighting where relevant).

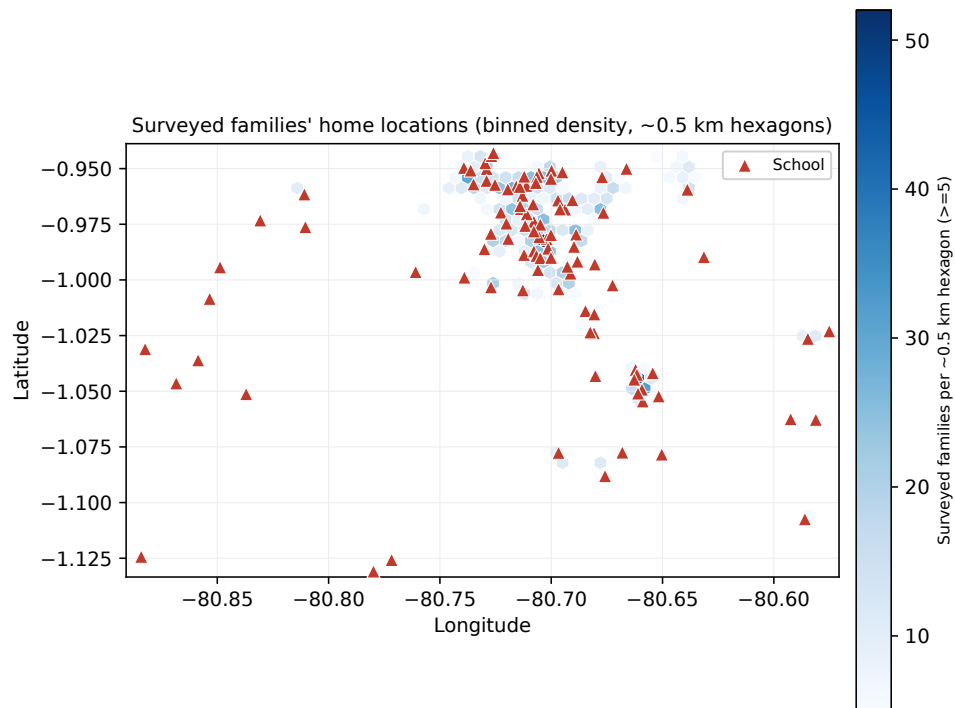


Figure 13: Home locations of surveyed families, shown as a binned density (~ 0.5 km hexagons; bins with fewer than five families suppressed) to avoid disclosing individual household locations. Red triangles are schools. The map shows the surveyed households only; the balance comparisons in Table 22 speak to selection on outcome-relevant margins, but within-canton geographic coverage is not itself tested.

Table 22: Representativeness: surveyed vs. non-surveyed applicants

	Surveyed	Not surveyed
Manta-origin (%)	70	61
Block deprivation, NBI (%)	42.6	47.8
Application-list length (mean)	2.05	1.77
True admission risk (%)	23.4	25.5
Placed in a listed school (%)	79	78

All standardized differences ≤ 0.17 ; balanced on grade. Block deprivation from the 2022 census manzana join.

I Detailed results

Information and list length. Families’ awareness of the alternatives they did not apply to is limited. Asked how well they know each of up to five nearby public schools outside their application, respondents answer “no la conozco” for **70.6%** of school–family pairs, “sólo de

nombre” for 21.6%, and “la conozco bien” for 7.7%.³¹ Lists are also short: **55% list a single school** (mean 2.05; on the in-market estimation sample the single-list share is 57.8/66.8/66.1% across Preschool 1/2 and Primary 1).

Why lists are short. Table 23 gives the stated reasons. The dominant reason is *scarcity of acceptable nearby options*, followed by *confidence* of placement; only about 6% of those answering cite any difficulty finding schools—the only response in this item that names an information friction. (Shares are shown over families who answered and over all respondents. The remaining 12.5% did not answer the item; 192 of those 234 are respondents who did not finish the instrument, so non-response here is concentrated among survey attrition rather than spread at random. Assigning every non-answerer to the information-friction category—the most adverse assumption available—would raise that share to 18.0% of all respondents, still well below the 48.1% who report that no acceptable public school is close enough.)

Table 23: Reason for not extending the application list

Reason	% of answering	% of all
No acceptable public school close enough (scarcity)	55.0	48.1
Confident of a seat in a listed school	33.1	29.0
Difficulty finding more schools (<i>information friction</i>)	6.3	5.5
Knows the alternatives, prefers to remain unassigned	4.1	3.6
Would enrol private if unassigned	1.5	1.3
(left blank)	—	12.5
<i>N</i>	1,639	1,873

Search at the margin where lists lengthen. A small group did extend the list they had started: 69 families (3.7% of respondents) added at least one school. Asked whether they had to look for more information to be persuaded to add them, 42 of the 64 who answered (65.6%) said they did. The group is small and self-selected—these are precisely the families for whom search was worth undertaking—so we do not read it as an aggregate statement. It does indicate that search is costly at the margin where lists lengthen, which is consistent with the limited knowledge of unlisted alternatives reported above and with our treating the discovery of options as a separate margin from acting on the preferences families already hold.

Beliefs and satisfaction. Families are confident and satisfied (Table 24): the median subjective probability of a seat at a chosen school is 98%, and 82% rate the process in the top band (16–20 of 20).

Were short lists and confidence justified? Linking the survey to the realized deferred-acceptance assignment (99.8% matched; beliefs by bucket in Table 25), short lists were

³¹The platform field `cantKnowledge` records how many such schools the instrument *displayed* to each family—capped at five, and equal to five for 92.3% of respondents—and is therefore a measure of local public-school density rather than of household knowledge. The knowledge items are `knowPub1–5`, reported above.

Table 24: Beliefs and satisfaction

Measure	Value
Median P(seat at a chosen school)	98%
share believing $\geq 80\%$	75%
Process rating 16–20 / 20	82%
rating 11–15	9%
rating 1–10	9%

Satisfaction is *pre-results* (process satisfaction), as the survey preceded the release of assignments.

vindicated *ex post* for most families: **72% of single-listers obtained their one school**, and realized placement in a listed school exceeds 78% at *every* stated-confidence level. This says that adding fallbacks would not have changed the outcome for those families; it does not establish that short lists were complete or ex ante optimal, since under a strategy-proof mechanism the 28% who missed their sole listed school could only have gained from naming acceptable alternatives.

Table 25: Survey beliefs and realized assignment

Stated P(seat) bucket	N	Mean stated	Realized placed in a listed school
$\leq 49\%$	103	27%	87.4%
50–79%	309	61%	79.3%
80–99%	436	87%	79.8%
100%	806	100%	78.7%
All matched respondents	1,869	—	79.8% (69.8% first choice)

Realized outcome under the deferred-acceptance assignment families experienced. The all-respondent figures mirror the paper’s DA aggregate (78.2% / 69.7%).

J Summary and takeaways

(1) Short lists are not a failure to locate alternatives. A majority list a single school, and the stated reason for not listing more is overwhelmingly scarcity of acceptable nearby options or confidence of placement; only $\sim 6\%$ cite difficulty finding more schools. Families’ knowledge of the alternatives they did not apply to is nonetheless thin (70.6% “no la conozco”). Scarcity and confidence are therefore the most commonly *reported* reasons, and few families name search difficulty; but since the item asked for a single principal reason and most unlisted neighbours were unknown, this does not establish that the omitted schools had been evaluated and found unacceptable. Short lists reflect a constrained acceptable set as families describe it, not a fully informed comparison.

(2) Beliefs are confident, and placement was high across the board. Median confidence is 98%, and realized placement in a listed school is high at every stated-confidence level. This is not a calibration result: the belief item asks about the *first-choice* school while the realized

outcome we can match is placement in *any* listed school, and uniformly high placement across belief categories shows high placement rather than accuracy.

(3) Outcomes vindicated most short lists ex post. For the 72% of single-listers who obtained their school, added fallbacks would have changed nothing; the 28% who missed theirs could only have gained from listing more. The matched sample reproduces the assignment’s aggregate placement rates.

(4) Implications for the welfare analysis. Limited awareness of unlisted alternatives cuts in both directions and we do not claim it signs the bias: it may attenuate the value of a preference-respecting mechanism, but because the distance rule assigns without regard to what a family knows, treating an unlisted school as an unwanted one also overstates the welfare forgone under that rule (see Section 6.2). Process satisfaction is consistent with, but does not independently quantify, the estimated gain: it was elicited before results were released and under the new mechanism only, so no distance-rule counterpart exists against which to compare it—direct survey evidence of a kind unavailable in the closest prior welfare analysis of coordinated assignment ([Abdulkadiroğlu et al., 2017](#)).